Class 6
MATHEMATICS
Volume 2
Solutions

Table of Content

7.	Algebra	1 – 17
8.	Geometry	18 – 39
9.	Trigonometry	40 – 56

7. ALGEBRAIC EXPRESSIONS

WORKSHEET-1

Level – I

1) A	2) C	3) C	4) C	5) B	6) D
7) B	8) A	9) B	10) A		

Level – II

1) A	2) D	3) B	4) D	5) A	6) C
7) C	8) A	9) A,B,C	10) C	11) A	12) A
13) B	14) A	15) B	16) A	17) C	18) B

1. Conceptual
2. Conceptual
3. Conceptual
4. Conceptual
5. Conceptual
6. $2x + 7 = 2(0) + 7 = 7$
7. $x^3 = (-2)^3 = -8$
8. Conceptual
9. Conceptual
10. $3 + 1 + 1 = 5$
11. Conceptual
12. $x + 2y + 3z = 3 + 2(2) + 3(1) = 10$
13. $m^n = 3^2 = 9$
14. Conceptual
15. $(1 - a)(1 + a + a^2) = (1 + a)(1 - a + a^2)$
    $1 - a^3 = 1 + a^3 \Rightarrow 2a^3 = 0 \Rightarrow a = 0$
16. Conceptual
17. $x - x^3 + x^5$
18. $s = \frac{3+4+5}{2} = 6;\ A = \sqrt{6(6-3)(6-4)(6-5)} = 6$

Level – III

19) A,C	20) D	21) D	22) B	23) C	24) A
25) A	26) a-q;b-s;c-p;d-r		27) C	28) C	29) 60
30)a-q;b-p,r;c-q;d-s					

19. Conceptual
20. $5\left(\frac{4}{5}\right) - \pi = 4 - \pi$
21. $\frac{4^4}{4} + \frac{4^3}{2} + \frac{4^2}{8} = 64 + 32 + 2 = 98$
22. $ax + b = 0 \Rightarrow x = -\frac{b}{a}$

23. $\frac{1+4+9}{4+6+6} = \frac{14}{16} = \frac{7}{8}$
24. $\sqrt{100(100-40)(100-75)(100-85)} = 1500$
25. Conceptual
26. Conceptual
27. $(-8)^2 + 2(-8)(3) + 3^2 = 64 - 48 + 9 = 25$
28. $(-10) - 8 + \frac{48}{15} - \frac{3}{4} = -\frac{311}{20}$
29. $a = 2, b = -1, c = -1$

$$\frac{2a^2}{3} + \frac{3\,b^3}{4} - \frac{4c^3}{5} - 2$$

$$\frac{2(4)}{3} + \frac{3(-1)}{4} - \frac{4(-1)}{5} - \frac{2}{1} \Rightarrow \frac{8}{3} - \frac{3}{4} + \frac{4}{5} - \frac{2}{1} \Rightarrow \frac{160 - 45 + 48 - 120}{60} \Rightarrow \frac{40+3}{60} = \frac{43}{60}$$

30. Conceptual

WORKSHEET – 2

Level – I

1) B	2) B	3) A	4) B	5) C	6) A	7) A
8) A	9) B	10) A				

Level – II

1) A	2) A	3) B	4) B	5) B	6) C	7) A
8) B	9) A	10) B	11) A	12) A	13) C	14) A
15) A	16) C	17) A				

SOLUTIONS

1. $a^2 + b^2 - (2a^2 + b^2) = -a^2$
2. $x + 3 + 6x = 7x + 3$
3. $(2(0) + 1) + (3(0) + 2(1)) = 1 + 2 = 3$
4. $3x^2 - y^2 + z^2 - x^2 - y^2 - z^2 \Rightarrow 2x^2 - 2y^2$
5. Conceptual.
6. $(2(1) + 2 + 3) + (5(1) - 2(2) + 3(3)) = 17$
7. $(2ab + bc - 3ca) + (5ab + 7bc + 9ca) = 7ab + 8bc + 6ca$
8. $(15x^2 + 4xy - 2b^2 - 4) + (4x^2 - 12xy - 2b^2 + 1) \Rightarrow 19x^2 - 8xy - 4b^2 - 3$
9. $(2x^2 + 2xy + 3y^2) + (5x^2 - 2xy + 3y^2) \Rightarrow 7x^2 + 6y^2$
10. $(4x^2 - 5xy + 3y^2) + (8x^2 + 7xy - 6y^2) + (-2x^2 - 9xy + 13y^2) \Rightarrow 10x^2 + 10y^2 - 7xy$
11. $(a^4 + 3a^2b^2 + b^4) - (a^4 - 2a^2b^2 + b^4) \Rightarrow 5a^2b^2$
12. $(x^3 - 8x^2y^2 + y^3) - (x^3 - 3x^2y^2 + y^3) \Rightarrow -5x^2y^2$
13. $\left(\frac{11}{2}xy + \frac{12}{5}y + \frac{13}{7}x\right) + \left(-\frac{11}{2}y - \frac{12}{5}x - \frac{13}{7}xy\right)$

$$\Rightarrow \frac{77-26}{14}xy + \frac{24-55}{10}y + \frac{65-84}{35}x \Rightarrow \frac{51}{14}xy - \frac{31}{10}y - \frac{19}{35}x$$

14. $\frac{7}{2} - \frac{x}{3} - \frac{x^2}{5} - \frac{9}{2} - \frac{x}{2} - \frac{3x^2}{5} - \frac{7x^3}{4} \Rightarrow -\frac{7x^3}{4} - \frac{4x^2}{5} - \frac{5x}{6} - 1$
15. $\frac{3}{2}a + \frac{2}{3}a + \frac{5}{3}a - \frac{5}{4}b - \frac{7}{2}b + \frac{5}{2}b + \frac{2}{5}c + \frac{7}{2}c - \frac{5}{4}c$

$$\Rightarrow a\left[\frac{3}{2} + \frac{2}{3} + \frac{5}{3}\right] + b\left[-\frac{5}{4} - \frac{7}{2} + \frac{5}{2}\right] + c\left[\frac{2}{5} + \frac{7}{2} - \frac{5}{4}\right] \Rightarrow \frac{23}{6}a - \frac{9}{4}b + \frac{53}{20}c$$

16. $A + B + C = \left(\frac{7}{2}x^3 + \frac{3}{2}x^3\right) + \left(-\frac{1}{2}x^2 + \frac{7}{4}x^2 + \frac{3}{2}x^2\right) + \left(-x - \frac{5}{2}x\right) + \frac{5}{3} - 2 + \frac{1}{3}$

$\Rightarrow x^3\left(\frac{7}{2}+\frac{3}{2}\right)+x^2\left(-\frac{1}{2}+\frac{7}{4}+\frac{3}{2}\right)+x\left(-1-\frac{5}{2}\right)+\frac{5-6+1}{3}$

$\Rightarrow 5x^3+\frac{11}{4}x^2-\frac{7x}{2}+\frac{0}{3}\Rightarrow 5x^3+\frac{11x^2}{4}-\frac{7x}{2}$

Let third side be " CA "

17. Given perimeter of triangle $= 7p^2-8p+9$
Perimeter of $\triangle ABC = AB+BC+CA$
$7p^2-8p-9 = 2p^2+1+11p^2-3p+5+CA$
$7p^2-8p+9 = 13p^2-4p+6+CA$
$CA = -6p^2-4p+3$

Level – III

18) B,C,D 19) A 20) B 21) A 22) A
23) a-r,b-s,c-t,d-q 24) B 25) A 26) C 27) A
28) A 29) A 30) a-t, b-s, c-s, d-p

18. $\left(\frac{4a+5a}{3}\right)+2a=\frac{9a}{3}+2a=3a+2a=5a$
so it is greater than a,2a,3a
19. Given that x = 2
$3x^2+x+a=17\Rightarrow 3(2)^2+2+a=17\Rightarrow 12+2+a=17\Rightarrow a=17-14\Rightarrow a=3$
20. $A+B=(4x^2+2x-3)+(-2x^2-2x+4)=2x^2+1$
21. $B-C=(-2x^2-2x+4)-(6x^2+9x-1)=-8x^2-11x+5$
22. $2A+A+B+C=3A+B+C=3(4x^2+2x-3)+(-2x^2-2x+4)+(6x^2+9x-1)$
$\Rightarrow 16x^2+13x-6$
23. a) $(x^2+x+2)+(-x^2-x-2)=0$
b) $(x^2+x+2)-(-x^2-x-2)=2x^2+2x+4$
c) $A-2(B)=(x^2+x+2)-2(-x^2-x-2)=3x^2+3x+6$
d) $2(x^2+x+2)+(-x^2-x-2)=x^2+x+2$
24. $(-x^3+3x^2+5x+4)-(-x^3+3x^2-5x+2)=+10x+2$
25. $3(2x+2y-2xy)=6x+6y-6xy$
26. $A=-x^2+2xy-y^2-x^2-2xy-y^2+x^2+2xy+y^2\Rightarrow -x^2+2xy-y^2$ is true
$B=3b^2+4c^2+9b^2+bc+2c^2-2b^2-2bc+c^2-c^2-2bc+b^2$
$\Rightarrow 11b^2+6c^2-3bc$ is false
27. $(6px-2qy+3rz)+(-6px+4qy-2rz)=2qy+rz$
28. $A-2B=(6px-2qy+3rz)-2(-6px-2rz+4qy)=18px-10qy+7rz$
29. $A+B+C=$
$(6px-2qy+3rz)+(-6px+4qy-2rz)+(4px-4qy-4rz)=4px-2qy-3rz$
30. a) $4x^3+3x^2-2x-1-4x^3+4x^2-6x+1-8x^3+4x^2+8x+6\Rightarrow -8x^3+11x^2+6$
b) $8x^3+3x^2-16x-6$
c) $8x^3+3x^2-16x-6$
d) $-16x^3+15x^2+8x+12$

WORKSHEET – 3

Level – I

1) A 2) A 3) B 4) A 5) C 6) C
7) C 8) C 9) C 10) B

Level – II

1) A	2) B	3) A	4) B	5) B	6) A
7) C	8) B	9) B	10) D	11) A	12) B
13) B	14) A	15) D	16) C	17) A	18) A
19) A	20) A	21) B	22) A	23) B	24) A
25) A,B	26) A	27) C	28) B	29) A	30) 3	31) A
32) B,D	33) A	34)a-t,b-p,c-s,d-q		35) -15	

SOLUTIONS

1. $(x^3 \times 2x) - (x^2 \times 2x) - (1 \times 2x) \Rightarrow 2x^4 - 2x^3 - 2x$
2. $(4 \times -3)(p^2 \times p^3)(q \times q^2) \Rightarrow -12p^5q^3$
3. $(-5 \times -10 \times 2)(a \times a^2 \times a^3) \Rightarrow 100a^6$
4. $(2 \times 4 \times 2)(\mathrm{x}^2 \times \mathrm{x} \times \mathrm{x}) \Rightarrow 16\mathrm{x}^4$
5. $\left(\frac{16}{9} \times \frac{9}{8}\right)(\mathrm{p}^3)(\mathrm{q})(\mathrm{r} \times \mathrm{r}) \Rightarrow 2\mathrm{p}^3\mathrm{qr}^2$
6. $(20 \times 10)(\mathrm{x} \times \mathrm{x})(\mathrm{y}^{20} \times \mathrm{y})(\mathrm{z} \times \mathrm{z}) \Rightarrow 200\mathrm{x}^2\mathrm{y}^{21}\mathrm{z}^2$
7. $(20xy \times 4xy) - (z \times 4xy) \Rightarrow 80x^2y^2 - 4xyz$
8. $\mathrm{A} \times \mathrm{B} = (10 \times 5)(\mathrm{x}^3 \times \mathrm{x}^2)\mathrm{y}^2 \Rightarrow 50\mathrm{x}^5\mathrm{y}^2$
9. $\mathrm{A} \times \mathrm{B} \times \mathrm{C} = (5 \times -5 \times 4)(\mathrm{x} \times \mathrm{x} \times \mathrm{x}^2) \Rightarrow 100x^4$
10. $\left(2 \times \frac{5}{3}\right)(p^2 \times p^4)(q^2 \times q^2)(z) \Rightarrow \frac{10}{3}p^6q^4z$
11. $(A + B)\mathrm{C} = (5x + 4x)5x \Rightarrow (9\mathrm{x})(5\mathrm{x}) = 45\mathrm{x}^2$
12. $(A - B)\mathrm{C} = (10\mathrm{a} - 4a)6x \Rightarrow (6\mathrm{a})(6x) = 36ax$
13. $(2x \times 4y) + (2\mathrm{a} \times 3b) \Rightarrow 8\mathrm{xy} + 6\mathrm{ab}$
14. 0
15. $(5 \times -10 \times -2.1)(\mathrm{a}^6 \times a \times a^2)(\mathrm{b}^2 \times \mathrm{b}^3) \Rightarrow 105(\mathrm{a}^9)(\mathrm{b}^5) \Rightarrow 105 \times 1^9 \times \left(\frac{1}{2}\right)^5 = \frac{105}{32}$
16. $\left(\frac{4}{3}a \times \frac{2}{4}b\right) + (6x \times 2) \Rightarrow \frac{2}{3}ab + 12x$
17. $(4\mathrm{a} \times 2\ \mathrm{b}) - (2x \times 4x) \Rightarrow 8ab - 8x^2$
18. $(0.2 \times 0.2 \times 0.2)(xyz) = 0.008xyz$
19. $(0.5x \times 2x) - (5.2y \times 3y) \Rightarrow x^2 - 15.6y^2$
20. $\frac{A \times B}{4} = \frac{(4x^2 \times 2x) - (2x \times 2x) - (8 \times 2x)}{4} \Rightarrow 2x^3 - x^2 - 4x$
21. $(2x \times 3x) + (2x \times 4y) = 6x^2 + 8xy$
22. $-120c^5d^4$
23. Conceptual
24. $k = \frac{6(\mathrm{x} - 2\mathrm{y})}{(x - 2y)} \Rightarrow k = 6$
25. $20a + 12b = 2(10a + 6b) = 4(5a + 3b)$
26. $(x - a)(x + b) = x^2 + kx - 54 \Rightarrow x^2 + (b - a)x - 54 = x^2 + kx - 54$
    $\Rightarrow b - a = k, ab = 54$
    $b = 9, a = 6$
    $\therefore \mathrm{k} = 9 - 6 = 3$
35. $(2x + 3)(x - 5) = 2x^2 - 7x + k$
    $2x^2 - 7x - 15 = 2x^2 - 7x + k$
    $-15 = k$
    $\therefore k = -15$

WORKSHEET – 4

Level – I

1.C	2.D	3.B	4.D	5.A	6.B	7.A

Level – II

1. C	2. C	3. C	4. A	5. B	6. C
7. B	8. D	9. B	10. A	11. B	12. C
13. C	14. B	15. C	16. A	17. D	18. C
19. B					

SOLUTIONS

1. $\left(\frac{a}{2}+3\right)^2=\left(\frac{a^2}{4}+2\times\frac{a}{2}\times 3+9\right)=\frac{a^2}{4}+3a+9$
2. $(2\mathrm{x}-1)^2=(4\mathrm{x}^2+1-4\mathrm{x})$
3. $(ab+5)(ab-5)=(ab)^2-5^2=a^2b^2-25$
4. $\left(x+\frac{1}{x}\right)^3=x^3+\frac{1}{x^3}+3\left(x\times\frac{1}{x}\right)\left(x+\frac{1}{x}\right)=x^3+\frac{1}{x^3}+3x+\frac{3}{x}$
5. $(x+y+2)^2=x^2+y^2+4+2(xy+2y+2x)$
6. $x-\frac{1}{x}=7\Rightarrow x^2+\frac{1}{x^2}=\left(x-\frac{1}{x}\right)^2+2\times x\times\frac{1}{x}=\left(x-\frac{1}{x}\right)^2+2\Rightarrow x^2+\frac{1}{x^2}=7^2+2=49+2=51$
7. $\mathrm{x}+\frac{1}{\mathrm{x}}=7\Rightarrow \mathrm{x}^2+\frac{1}{\mathrm{x}^2}=\left(\mathrm{x}+\frac{1}{\mathrm{x}}\right)^2-2\times \mathrm{x}\times\frac{1}{\mathrm{x}}=\left(\mathrm{x}+\frac{1}{\mathrm{x}}\right)^2-2$
   $\Rightarrow \mathrm{x}^2+\dfrac{1}{\mathrm{x}^2}=7^2-2=49-2=47$
8. $(2x-3y+z)^2=(2x)^2+(-3y)^2+(z)^2+2(2x\times -3y+(-3y)\times z+z\times 2x)$
   $=4\mathrm{x}^2+9\mathrm{y}^2+\mathrm{z}^2+2(-6\mathrm{xy}-3\mathrm{yz}+2\mathrm{xz})$
9. $(2a+3b)^2+(2a-3b)^2=4a^2+9b^2+12ab+4a^2+9b^2-12ab=8a^2+18b^2$
10. $a^3+b^3=(a+b)^3-3ab(a+b)\Rightarrow 8a^3+27b^3=(2a+3b)^3-3(2a)(3b)(2a+3b)$
    $\Rightarrow (2a+3b)^3-18ab(2a+3b)=9^3-18\times 3\times 9=729-486=243$
11. $\mathrm{a}-\mathrm{b}=8,\mathrm{ab}=15\Rightarrow \mathrm{a}^2+\mathrm{b}^2=(\mathrm{a}-\mathrm{b})^2+2\mathrm{ab}=8^2+30=64+30=94\mathrm{c}$
12. $(a+b)^2=a^2+b^2+2ab=200+2(48)=296$
13. $x+y+z=0\Rightarrow x+y=-z$
    cubing on both sides
    $x^3+y^3+3xy(x+y)=-z^3$
    $x^3+y^3+3xy(-z)=-z^3$
    $x^3+y^3+z^3=-3xy(-z)$
14. $a+b=11,a^2+b^2=61$
    squaring on both sides
    $(a+b)^2=11^2,\Rightarrow a^2+b^2+2ab=121\Rightarrow 61+2ab=121\Rightarrow 2ab=60\Rightarrow ab=30$
15. $4x^2+\frac{9}{x^2}+12, x=2=4(2)^2+\frac{9}{2^2}+12=\frac{64+9+48}{4}=\frac{121}{4}$
16. $(-2x-3y-4z)^2=4x^2+9y^2+16z^2+2(-2x\times -3y+-3yx-4z+-4z\times -2x)$
    $=4\mathrm{x}^2+9\mathrm{y}^2+16\mathrm{z}^2+2(6\mathrm{xy}+12\mathrm{yz}+8\mathrm{xz})$
17. $81a^2+9b^2+54ab, a=2, b=4=81(4)+9(16)+54(2)(4)=164+144+432=900$

18. $x+\frac{1}{x}=5$

squaring on both sides

$x^2+\frac{1}{x^2}+2=25 \Rightarrow x^2+\frac{1}{x^2}=23$

squaring on both sides

$x^4+\frac{1}{x^4}+2=529 \Rightarrow x^4+\frac{1}{x^4}=527$

19. $(x+1)(x-1)(x+2)(x-2)=(x^2-1)(x^2-4)=x^4-5x^2+4$

Level – III

20. B, C 21.A, B 22. A 23. A 24. B 25.B
26. a-s;b-r;c-q;d-p 27. A, C 28. B 29. D 30.B
31. a-p;b-q;c-s;d-t 32.C 33. A 34. A

20. $\left(\frac{2}{3}x+\frac{1}{2}y\right)^2=\left(\frac{2}{3}x\right)^2+2\left(\frac{2}{3}x\right)\times\left(\frac{1}{2}y\right)+\left(\frac{1}{2}y\right)^2=\frac{4x^2}{9}+\frac{2xy}{3}+\frac{y^2}{4}$
21. $(p-2q)^2=p^2+4q^2-4pq=(p-2q)(p-2q)$
22. I. $(9x+y)^2=81x^2+18xy+y^2$
II. $(a+b)^2=a^2+2yb+b^2$ both true
23. $(100+2)(100-2)=(100)^2-(2)^2=10000-4=9996$
24. $(x-2y)^2=x^2-4xy+4y^2$
25. $(2a+3\ b)^2=4a^2+9\ b^2+12ab$
26. (a) $(x+2)(x+3)=x^2+5x+6$
(b) $(x+1)(x+2)=x^2+3x+2$
(c) $(x+y)^2-4xy=x^2+y^2+2xy-4xy=x^2+y^2-2xy=(x-y)^2$
(d) $(x-y)^2+4xy=x^2+y^2-2xy+4xy=x^2+2xy+y^2=(x+y)^2$
27. $(2x-3y)^2=(2x-3y)(2x-3y)=4x^2-12xy+9y^2$
28. $(a^2+b^2)^2=(a^2)^2+(b^2)^2+2a^2b^2=a^4+b^4+2a^2b^2$
29. $(a^2-b^2)^2=(a^2)^2+(b^2)^2-2a^2b^2=a^4+b^4-2a^2b^2$
30. $(a^2+b^2)^2+(a^2-b^2)^2=a^4+2a^2b^2+b^4+a^4+b^4-2a^2b^2=2(a^4+b^4)$
31. a) $x^2-4x+\cdots=x^2-4x+2^2=(x-2)^2$
b) $y^2-8yz+(2z)^2=(xy-2z)^2$
c) $p^2+\cdots+4q^2=p^2+2\times p\times 2q+(2q)^2$
d) $41^2+\cdots+9\ m^2=(21)^2+2(21)(3\ m)+(3\ m)^2$

32. $x-\frac{1}{x}=5$

$x^2+\frac{1}{x^2}=\left(x-\frac{1}{x}\right)^2+2=5^2+2=25+2=27$

33. $x^4+\frac{1}{x^4}=\left(x^2-\frac{1}{x^2}\right)^2-2=27^2-2=729-2=727$

34. $\left(x-\frac{1}{x}\right)^3=x^3-\frac{1}{x^3}-3x\times\frac{1}{x}\left(x-\frac{1}{x}\right)$

$\Rightarrow x^3-\frac{1}{x^3}=\left(x-\frac{1}{x}\right)^3+3x\times\frac{1}{x}\left(x-\frac{1}{x}\right)\Rightarrow 5^3+15=125+15=140$

WORKSHEET – 5

Level – I

1. A	2. C	3. A	4. A	5. B	6. D
7. B	8. A	9. B			

Level – II

1. A	2. C	3. B	4. A	5. A	6. A
7. B	8. D	9. C	10. C	11. A	12. B
13. B	14. B	15. A			

SOLUTIONS

1. Conceptual
2. Conceptual
3. $100I = PNR \Rightarrow\therefore P = \frac{100I}{NR}$
4. $b = \frac{2A}{h}$
5. $M = \frac{2E}{v^2}$
6. $I = \frac{PNR}{100} \Rightarrow N = \frac{100I}{PR} \Rightarrow \text{N} = \frac{100\times250}{750\times\frac{15}{4}} = 8\frac{8}{9}$ years
7. $l = \frac{V}{bh} \Rightarrow \frac{10^5}{} = \frac{5}{4}$ cm
8. $\text{A} = s^2 \Rightarrow \text{s} = \sqrt{A} = \sqrt{169} = 13$ units.
9. $\text{t} = \sqrt{\frac{2\text{ s}}{\text{g}}} = \sqrt{\frac{2\times625}{5}} = \sqrt{25\times10}$
10. Conceptual
11. $5(F-32) = 9C \Rightarrow F - 32 = \frac{9C}{5} \Rightarrow F = \frac{9C}{5} + 32$
12. $2a = \frac{2s}{n} - (n-1)d \Rightarrow a = \frac{2s-n(n-1)d}{2n} \Rightarrow \frac{2\times185-10(10-1)3}{2\times10} = 5$
13. $v = \pi(R^2 - r^2)h \Rightarrow \text{h} = \frac{v}{\pi(R^2-r^2)} \Rightarrow \text{h} = 25$
14. $s = ut - \frac{1}{2}gt^2 \Rightarrow 10 = 500 - \frac{1}{2}g(100) \Rightarrow 50g = 490 \Rightarrow g = \frac{490}{50} = 9.8$
15. $4\emptyset^{20} =$

Level – III

16.A, B, C, D	17. A	18. B	19. B	20. A
21.a-r;b-p;c-q;d-s	22.10	23.B	24.D	25.a-q;b-r;c-p;d-q
26. A	27. A	28. B		

16. $\text{V} = l\text{bh} \Rightarrow l = \frac{v}{bh} \Rightarrow b = \frac{v}{lh} \Rightarrow h = \frac{v}{lb}$
17. Cost price = Rs xy selling price = Rs xz
    $\text{P} = \text{SP} - \text{CP} \Rightarrow \text{Rs}(\text{xz} - \text{xy})$ St-I: (1) is true
18. $c = \frac{5}{9}(F-32) \Rightarrow \frac{9c}{5} = F - 32 \Rightarrow F = \frac{9c}{5} + 32$
19. $F = \frac{9\times100}{5} + 32 \Rightarrow \text{F} = 212^\circ$
20. $0 = \frac{5}{9}(\text{ F} - 32) \Rightarrow \text{F} = 32^\circ$
21. Conceptual
22. $h = \frac{2A}{b}, h = \frac{2(2\emptyset)^5}{4} = 10$ cm

23. Conceptual
24. $625 = 100 + 100 \times 5r \Rightarrow r = \frac{21}{20}$
25. Conceptual
26. $340 = \left(\frac{100+\frac{25}{4}}{100}\right)C \Rightarrow 340 = \frac{425}{400}\text{C} \Rightarrow \text{C} = \frac{340\times400}{425} = 320$
27. $\text{S} = \left(\frac{100+\frac{25}{2}}{100}\right)600 = \text{Rs. } 675$
28. $720 = \left(\frac{100+\text{g}}{100}\right)600 \Rightarrow \text{g} = 20\%$

WORKSHEET – 6

Level – I

1) A	2) A	3) D	4) A	5) C	6) D
7) A	8) B	9) A	10) B		

Level – II

1) C	2) C	3) B	4) C	5) A	6) B
7) B	8) C	9) C	10) D	11) A	12) C
13) C	14) C	15) A	16) B	17) A	18) C
19) C	20) B	21) B	22) C		

SOLUTIONS

1. $2\text{y} - 3\text{y} + 3\text{ m} \Rightarrow -\text{y} + 3\text{x}$
2. $3\text{x} - 6\text{x} + 2\text{y} - \text{x} \Rightarrow -4\text{x} + 2\text{y}$
3. $-\text{p} - \{2\text{q} - 6\text{p} - 4\text{q}\} \Rightarrow -\text{p} - \{-2\text{q} - 6\text{p}\} \Rightarrow -\text{p} + 2\text{q} + 6\text{p} \Rightarrow 5\text{p} + 2\text{q}$
4. $\text{A} = \frac{16\text{x}+8}{4} = 4\text{x} + 2,\ \text{B} = \frac{15\text{x}-10}{-5} = 3\text{x} + 2$
   $A - B = 4x + 2 - (-3x + 2) \Rightarrow 4x + 3 + 3x - 2 = 7x$
5. $\frac{3}{4}(x + 1) - \frac{1}{2} - x + 1 = \frac{3}{4}x + \frac{3}{4} - \frac{1}{2} - x + 1 = \frac{3x+3-2-4x+4}{4} = \frac{-x+5}{4}$
6. $\text{a} + \text{b} - \text{c} + \text{b} + \text{c} - \text{a} + \text{c} + \text{a} - \text{b} = \text{a} + \text{b} + \text{c}$
7. $\text{A} = \text{x}^2 - \text{y}^2 + \text{z}^2,\ \text{B} = \text{y}^2 - \text{z}^2 + \text{x}^2, \text{C} = \text{z}^2 - \text{x}^2 + \text{y}^2$
   $\text{A} + \text{B} + \text{C} = \text{x}^2 + \text{y}^2 + \text{z}^2$
8. $P + Q = 13a - 14b, P - Q = -17a - 6b$
   $(\text{P} + \text{Q}) + (\text{P} - \text{Q}) = 13\text{a} - 14\text{ b} - 17\text{a} - 6\text{ b} = -4\text{a} - 20\text{ b} = -4(\text{a} + 5\text{ b})$
9. $7\text{x} - [8\text{y} - \{6\text{x} - (2\text{z} - 7\text{y}) + 5z - 4(2\text{x} - 7\text{y} - 5\text{z})\}]$
   $= 7\text{x} - [8\text{y} - (6\text{x} - 2\text{z} + 7\text{y} + 5\text{z} - 8\text{x} + 28\text{y} + 20\text{z}\}], = 7\text{x} - [8\text{y} - \{-2\text{x} + 35\text{y} + 23\text{z}\}]$
   $= 7\text{x} - [8\text{y} + 2\text{x} - 35\text{y} - 23\text{z}] \Rightarrow 7\text{x} + 27\text{y} - 2\text{x} + 23\text{z},\ \Rightarrow 5\text{x} + 27\text{y} + 23\text{z}$
10. $(\text{x}^2 - 4)(\text{x}^2 + 4)(\text{x}^4 + 16) \Rightarrow (\text{x}^4 - 16)(\text{x}^4 + 16) \Rightarrow \text{x}^8 - 256$
11. $\text{A} - \text{B} - \text{C} + 2\text{ B} \Rightarrow \text{A} - \text{B} + \text{C} + 2\text{ B} \Rightarrow \text{A} + \text{B} + \text{C}, \text{x} - \text{y} + \text{xy} + \text{y} - \text{z} + \text{yz} + \text{z} - \text{x} + \text{zx}$
    $\text{xy} + \text{yz} + \text{zx}$
12. $\text{P} - \{\text{Q} - (\text{P} - \text{Q} - \overline{\text{P} - \text{Q}})\} = \text{P} - \{\text{Q} - (\text{P} - \text{Q} - \text{P} + \text{Q})\}$
    $\text{P} - \text{Q} \Rightarrow (2\text{x} + \text{y})(5\text{x} - 3\text{y}) - (\text{x} + \text{y})(5\text{x} - 3\text{y}), (5\text{x} - 3\text{y})(2\text{x} + \text{y} - \text{x} - \text{y}), \text{x}(5\text{x} - 3\text{y})$
13. $\{(\text{x} - 5)(\text{x} + 2)\} + 3\text{x} + 10 \Rightarrow \{\text{x}^2 - 5\text{x} + 2\text{x} - 10\} + 3\text{x} + 10, \Rightarrow \{\text{x}^2 - 3\text{x} - 10\} + 3\text{x} + 10$
    $\Rightarrow \text{x}^2 - 3\text{x} - 10 + 3\text{x} + 10,\ \Rightarrow \text{x}^2$
14. $8\text{a} - [9\text{ b} - \{8\text{a} - (2\text{ b} - \overline{\text{a} + 6}\text{ b})\}]$
    $8\text{a} - [9\text{ b} - \{8\text{a} - (2\text{ b} - \text{a} - 6\text{ b})\}] \Rightarrow 8\text{a} - [9\text{ b} - \{8\text{a} - 2\text{ b} + \text{a} + 6\text{ b}\}]$

$\Rightarrow 8a-[9\ b-8a+2\ b-a-6\ b]\Rightarrow 8a-[5\ b-9a],\ \Rightarrow 8a-5\ b+9a\Rightarrow 17a-5\ b$

15. $A=x^2-5x-36,\ B=6x^2-x-12,$
$B-A=5x^2+4x+24, B+A=7x^2-6x-48$
$4(B+A)-3(B-A)=4(7x^2-6x-48)-3(5x^2+4x+24)$
$\Rightarrow 28x^2-24x-192-15x^2-12x-72,\Rightarrow 13x^2-36x-264$

16. $\{5(3x-4)-74+3x\}+48,\Rightarrow\{15x-20-74+3x\}+48\Rightarrow 15x-20-74+3x+48\Rightarrow 18x-46$

17. $\{(p+q)(p-q)+q^2)\div P,\{p^2-q^2+q^2\}\div P\Rightarrow\frac{p^2}{p}=p$

18. $A=5x-3y, B=11y, A+[2\ A-(A-B)]\Rightarrow A+[2\ A-A+B]$
$\Rightarrow A+A+B\Rightarrow 2\ A+B\Rightarrow 2(5x-3y)+11y,\Rightarrow 10x-6y+11y,\Rightarrow 10x+5y$

19. $P=9a-10\ b, Q=a+3\ b, P+Q-[-\{(P-Q)-(P+Q)\}]\Rightarrow P+Q-[-\{P-Q-P-Q\}]$
$\Rightarrow P+Q-[-P+Q+P+Q]\Rightarrow P+Q-2Q\Rightarrow P-Q, 9a-10\ b-a-3\ b=8a-13\ b$

20. $\{(a-3b+2c)+(-4a+9b-11c)\}-3a+4b+7c$
$=\{a-3b+2c-4a+9b-11c\}-3a+4b+7c,$
$\Rightarrow a-3\ b+2c-4a+9\ b-11c-3a+4\ b+7c,\ =-6a+10\ b-2c$

21.
$A=9x^2-9x-3, B=3x-6, B-\{-(A+3B)\}=B-\{-A-3B\}\Rightarrow B+A+3B$
$\Rightarrow A+4B$
$9x^2-9x-3+4(3x-6)\Rightarrow 9x^2-9x-3+12x-24,\ \Rightarrow 9x^2+3x-27$

22. $\frac{1}{2}x-\frac{1}{4}y+\frac{3}{5}z+\left(\frac{3}{2}x-\frac{5}{4}y+\frac{7}{5}z+2x-\frac{2}{4}y+z\right)$
$\Rightarrow\frac{1}{2}x-\frac{1}{4}y+\frac{3}{5}z+\frac{3}{2}x-\frac{5}{4}y+\frac{7}{5}z+2x-\frac{1}{2}y+z$
$\frac{1x+3x+4x}{2}+\frac{-1y-5y-2y}{4}+\frac{3z+7z+5z}{5}\Rightarrow\frac{8x}{2}-\frac{8y}{4}+\frac{15z}{5}\Rightarrow 4x-2y+3z$

Level – III

23) B,C	24) C,D	25) D	26) A	27) C	28) A
29) a-p; b-t; c-r; d-q		30) 2b	31) A,C	32) A,D	33) C
34) B,C	35) A,D	36) a-s,t; b-p; c-q; d-r			

23. $P=2x-[3y-\{4z-(2x-3y)\}],$
$P=2x-[3y-\{4z-2x+3y\}]\Rightarrow 2x-(3y-4z+2x-3y)$
$\Rightarrow 2x-3y+4z-2x+3y\Rightarrow 4z$

24. $5a-[9a-\{2a-2(2a-3b)-3(b-3b-2a)\}]$
$=5a-[9a-\{2a-4a+6b-3b+9b+6a\}]$
$=5a-[9a-2a+4a-6b+3b-9b-6a]$
$=5a-9a+2a-4a+6b-3b+9a+6a=-4a-4a+8a+12b=12b$

25. CLEARLY ST-2 IS TRUE, now, $8a-[5a-\{a-3(a-b)+2(a-\overline{b+a})\}]$
$8a-[5a-\{a-3a+3\ b-2\ b\}]=8a-[5a-\{-2a+b\}]\Rightarrow 8a-[5a+2a-b]$
$\Rightarrow 8a-[7a-b]\Rightarrow 8a-7a+b\Rightarrow a+b$

26. $A=a-2\ b-[4a-6\ b-\{3a-(+(5a-2\ b-\overline{3a-c+2\ b})\}]$
$=a-2b-[4a-6b-\{3a-c+(5a-2b-3a+c-2b)\}]$
$=a-2b-[4a-6b-\{3a-c+2a-4b+c\}]$

$= a - 2b - [4a - 6b - \{5a - 4b\}] \Rightarrow 2a$

27. $B = [4a - \{2a - 3b - (4a - 8b)\}] B = [4a - \{2a - 3b - 4a + 8b\}] \Rightarrow B = [4a - \{-2a + 5b\}]$
$B = [4a + 2a - 5b] \Rightarrow B = 6a - 5b$
28. $3(\mathrm{B} - \mathrm{A}) + 4(\mathrm{A} + \mathrm{B}) \Rightarrow 3\,\mathrm{B} - 3\,\mathrm{A} + 4\,\mathrm{A} + 4\,\mathrm{B} \Rightarrow \mathrm{A} + 7\,\mathrm{B}$
$2\mathrm{a} + 7(6\mathrm{a} - 5\,\mathrm{b}) \Rightarrow 2\mathrm{a} + 42\mathrm{a} - 35\,\mathrm{b}, \Rightarrow 44\mathrm{a} - 35\,\mathrm{b}$
29. (a) $3\mathrm{x} + [\mathrm{y} - \{2\mathrm{z} + (2\mathrm{x} - \overline{\mathrm{y} + \mathrm{z}})\}] \Rightarrow 3\mathrm{x} + [\mathrm{y} - \{2\mathrm{z} + (2\mathrm{x} - \mathrm{y} - \mathrm{z})\}]$
$3\mathrm{x} + [\mathrm{y} - \{2\mathrm{x} - \mathrm{y} + \mathrm{z}\}] = \mathrm{x} + 2\mathrm{y} - \mathrm{z}$, (b) $- [-2\mathrm{x} - \{3\mathrm{y} - (2\mathrm{x} - 3\mathrm{y}) + (3\mathrm{x} - 2\mathrm{y})\} + 2\mathrm{x}]$
$= \{3y - 2x + 3y + 3x - 2y\} = x + 7y,$
(c) $-(-(-(-(\mathrm{x})))) - (-(-(\mathrm{y}))) = \mathrm{x} - \mathrm{y}$
(d) $-5\mathrm{x} - [3\mathrm{y} - \{2\mathrm{x} - (2\mathrm{y} - \mathrm{x})\}] = -5\mathrm{x} - [3\mathrm{y} - \{2\mathrm{x} - 2\mathrm{y} + \mathrm{x}\}]$
$= -5x - [3y - \{3x - 2y\}]$
$= -5\mathrm{x} - 5\mathrm{y} + 3\mathrm{x} \Rightarrow -2\mathrm{x} - 5\mathrm{y}$
30. $6\mathrm{a} - [3\mathrm{a} + \{4\,\mathrm{b} - 2\mathrm{a} + \mathrm{b} + 5\mathrm{a}\} - 7\,\mathrm{b}], 6\mathrm{a} - [3\mathrm{a} + \{5\,\mathrm{b} + 3\mathrm{a}\} - 7\,\mathrm{b}] \Rightarrow 2\,\mathrm{b}$
31. $\mathrm{A} = 48\mathrm{a}^3\,\mathrm{b}^2$, $\mathrm{B} - \mathrm{C} = (2\mathrm{a} - 3\,\mathrm{b}) - (3\mathrm{a} - 5\,\mathrm{b}) 2\mathrm{a} - 3\,\mathrm{b} - 3\mathrm{a} + 5\,\mathrm{b} \Rightarrow -\mathrm{a} + 2\,\mathrm{b}$
$A \times (B - C) = 48a^3b^2(-a + 2b) \Rightarrow -48a^4 6^2 + 96a^3b^3$
32. $[\{\overline{7 - 2x} + 3y - (3x - y)\}\overline{-x - 5y + 11}] + 1, = [\{7 - 2x + 3y - 3x + y\} - x + 5y - 11] + 1$
$= [7 - 2\mathrm{x} + 3\mathrm{y} - 3\mathrm{x} + \mathrm{y} - \mathrm{x} + 5\mathrm{y} - 11] + 1, = 7 - 2\mathrm{x} + 3\mathrm{y} - 3\mathrm{x} + \mathrm{y} - \mathrm{x} + 5\mathrm{y} - 11 + 1$
$= -6x + 9y - 3$
5k 3k
33. $\frac{\overline{8 \quad 5}}{\frac{\mathrm{k}}{4}} \Rightarrow \mathrm{k}\left(\frac{25-24}{8\times5}\right)\times\frac{9}{\mathrm{k}} = \frac{1}{10}$ so, st – I is true

st $-2: \frac{7}{30}$ of $\left(\frac{7}{3} - \frac{7p}{15}\right) = \frac{7}{30}\left(\frac{7}{3} - \frac{7p}{15}\right) = \frac{49}{90}\left(1 - \frac{p}{5}\right) \frac{49(5-p)}{450}$, st II is false

34. $\frac{2x^3}{3} + \frac{6x^2}{7} + 2x - 3 - \frac{4x^3}{3} + \frac{4x^2}{7} + 5, \Rightarrow \frac{2x^3}{3} - \frac{4x^3}{3} + \frac{6x^2}{7} + \frac{4x^2}{7} + 2x - 3 + 5$
$\Rightarrow \frac{-2\mathrm{x}^3}{3} + \frac{10\mathrm{x}^2}{7} + 2\mathrm{x} + 2$
35. $\left(\frac{\mathrm{m}}{2} \times \frac{7}{3}\right) \div \left(\frac{9}{2} \times \frac{\mathrm{p}}{3}\right) = \frac{7\,\mathrm{m}}{6} \times \frac{6}{9\mathrm{p}} = \frac{7\,\mathrm{m}}{9\mathrm{p}}$
36. (a) $\mathrm{x} - [2\mathrm{y} + \{7 - (2\mathrm{x} - \mathrm{y})\}] = \mathrm{x} - [2\mathrm{y} + \mathrm{z} - 2\mathrm{x} + \mathrm{y}] = \mathrm{x} - 3\mathrm{y} - \mathrm{z} + 2\mathrm{x}, = 3\mathrm{x} - 3\mathrm{y} - \mathrm{z}$
(b) $9\mathrm{x} - [2\mathrm{y} - \{2\mathrm{x} + (\mathrm{x} - \mathrm{y} - \mathrm{x})\}] = 4\mathrm{x} - [2\mathrm{y} - \{2\mathrm{x} + \mathrm{x} - \mathrm{y} + \mathrm{x}\}] = 4\mathrm{x} - [2\mathrm{y} - \{4\mathrm{x} - \mathrm{y}\}]$
$= 4\mathrm{x} - 2\mathrm{y} + 4\mathrm{x} - \mathrm{y} = 8\mathrm{x} - 3\mathrm{y}$
(c) $5\mathrm{x} - [6\mathrm{x} - \{9\mathrm{x} - (10\mathrm{x} - 4\mathrm{x} + 3\mathrm{x})\}] \Rightarrow 5\mathrm{x} - [6\mathrm{x} - \{9\mathrm{x} - 9\mathrm{x}\}] \Rightarrow 5\mathrm{x} - [6\mathrm{x}] = -\mathrm{x},$
(d) $-1[\mathrm{x} - 3\{\mathrm{y} - 4(\mathrm{x} - \mathrm{y} - 8) + 4\mathrm{x}\} + 10]$
$= -[\mathrm{x} - 3\{\mathrm{y} - 4(\mathrm{x} - \mathrm{y} - 8) + 4\mathrm{x}\} + 10] = -[\mathrm{x} - 3\{5\mathrm{y} - 32\} + 10] = -\mathrm{x} + 15\mathrm{y} - 10\,\mathrm{b}$

WORKSHEET – 7

Level – I

1) B	2) A	3) A	4) A	5) A	6) C
7) C	8) D	9) B	10) A		

Level – II

1) D	2) A	3) D	4) A	5) D	6) C
7) D	8) B	9) A	10) C	11) D	12) B
13) A	14) A	15) D	16) A	17) A	18) C

SOLUTIONS:

1. $\frac{15a^3b}{5ab} = 3a^2$
2. $\frac{a^9}{a^8} = a$
3. $\frac{abc}{ac} = b$
4. $\frac{ab}{2a} = \frac{b}{2}$
5. $\frac{a^3}{a^5}$ $\because \frac{a^n}{a^m} = \frac{1}{a^{m-n}} \Rightarrow \frac{1}{a^{5-3}} = \frac{1}{a^2}$

6. $a^3b^4 \div a^2b = ab^3$ 7. $\frac{-33abc^2}{-33abc} = c$ 8. $a^3b^7 \div a^4b^6 = \frac{b}{a}$

9. $\frac{5a^3b-7ab^3}{ab} = \frac{ab(5a^2-7b^2)}{ab} = 5a^2 - 7b^2$ 10. $\frac{4x^2y^3}{4x^6y} = \frac{y^2}{x^4}$

10. Conceptual
11. $\frac{5x^2y^2(-2x+3y-1)}{5x^2y^2} = -2x + 3y - 1$
12. $\frac{x^4y-2x^3y^2+x^5y^5}{x^2y} = x^2 - 2xy + x^3y^4$
13. $\frac{7a^2b^3}{21ab^4} = \frac{a}{3b}$
14. $\frac{2x^2(8x^3+4x^2+2x+3)}{Ax} = 8x^3 + 4x^2 + 2x + 3$

    $Ax = 2x \Rightarrow A = 2 \Rightarrow \frac{8x^3 + 4x^2 + 10x}{2x} = 4x^2 + 2x + 5$
16. $\frac{A \times B}{4x^3y^3} = \frac{32x^{10}y^5+16x^6y^4+4x^4y^5}{4x^3y^3} = 8x^7y^2 + 4x^3y + xy^2$
17. $x + 2)x^2 + 4x + 4(x + 2$

    $x + 2)x^2 + 4x + 4(x + 2$

    $x^2 + 2x$ (−) (−)

    $2x + 4$

    $2x + 4$

    (0)

    $\therefore \frac{A}{C+D} = x + 2$

    $\therefore \frac{A+B}{C+D} = x + 2$ $2x + 4$ $2x + 4$
18. $\frac{X-Y=20x^4+28x^3+12x^2-20x}{(X+Y)(X-Y)=-16x^4-24x^3-8x^2+16x}$

Level – III

19) C	20) C	21) A	22) D	23) A	24) A
25) B	26) a-r;b-s;c-p;d-q		27) 2	28) 4	29) A,B,C,D
30) C	31) D	32) A			

19. $\frac{xy}{x^6} = \frac{y}{x^5}$
20. $\frac{a}{a^4}$ $\because \frac{a^n}{a^m} = \frac{1}{a^{m-n}} \Rightarrow \frac{1}{a^{1-4}} = \frac{1}{a^3}$
21. I: $\frac{15x^4y^3}{9xy^2} = \frac{5x^3y}{3}$ II: $\frac{a^5}{a^3} = a^2$
22. Conceptual

23. Conceptual
24. Conceptual
25. $\frac{20x^2y^6}{5x^8y^2} = \frac{4y^4}{x^4}$ 26. Conceptual
27. $\frac{px^3}{8x} = 3x^2 \Rightarrow px^3 = 24x^3 \Rightarrow p = 24 \Rightarrow \frac{qx^3}{4x} = 12x^2 \Rightarrow qx^3 = 48x^3 \Rightarrow q = 48$
    $\therefore \frac{qx^4y^3}{px^2y^2} = \frac{48x^4y^3}{24x^2y^2} = 2x^2y \Rightarrow Kx^2y = 2x^2y \Rightarrow K = 2$
28. 4
29. $\frac{2m^2+4m^3+2m^4-m^6}{\sqrt{2}m^2} = \frac{2m^2}{\sqrt{2}m^2} + \frac{4m^3}{\sqrt{2}m^2} + \frac{2m^4}{\sqrt{2}m^2} - \frac{m^6}{\sqrt{2}m^2} = \sqrt{2} + 2\sqrt{2}m + \sqrt{2}m^2 - \frac{1}{\sqrt{2}}m^4$
30. $\frac{a+b}{c} = \frac{a+c}{b} = \frac{b+c}{a} = k + 1 \Rightarrow (k+1)^3 \Rightarrow 2^3 = 8$
31. $\Rightarrow (k+1)^3 \Rightarrow 6^3 = 216\ 32 \cdot (k+1)^3$

WORKSHEET – 8

Level – I

1) A	2) C	3) A	4) B	5) D	6) C
7) A	8) D	9) B	10) A	11) B	

Level – II

1) B	2) C	3) D	4) B	5) C	6) A
7) C	8) B	9) C	10) A	11) B	12) A
13) C	14) B	15) A	16) A	17) B	18) C
19) C	20) A	21) C			

SOLUTIONS:

1. $14xyz = 2 \times 7 \times x \times y \times z$
   $28yz^3 = 2 \times 2 \times 7 \times y \times z \times z$
   $56xy^2z = 2^2 \times 2 \times 7 \times yyz \times 2 \Rightarrow 14yz$
2. $4a^2\ b^3 = 2 \times 2 \times a^2 \times b \times b^2, -12a^3\ b = -2 \times 2 \times 3 \times a^2 \times a \times b$
   $18a^4b^3 = 2 \times 3 \times 3 \times a^2 \times a^2 \times b \times b^2$
   $HCF = 2a^2\ b$
3. Given $4a^2\ b = 2 \times 2 \times a \times a \times b,\ 6a^2\ b^2 = 2 \times 3 \times a \times a \times b \times b$
   $3ab = 3 \times a \times b$
   $L.C.M = 2 \times 2 \times 3 \times a \times a \times b \times b = 12a^2\ b^2$
4. Given $2xyz, 3xz, 4xyz$
   $H.C.F = xz$
5. Given $17a^2\ b^4k^2, 51a^4k^4 \Rightarrow 17 \times 3a^4k^4\ \ \therefore LCM = 51a^4\ b^4k^4$
6. Given $a^3\ b^2,\ b^2c^4, a^3\ b^5c^4\ \ \therefore L.C.M = a^3\ b^5c^4$
7. Given $= 15x^3y^4 = 3 \times 5x^2y^4x$
   $20k^2x^2y^3 = 4 \times 5k^2x^2y^3$
   $30kx^3 = 3 \times 2 \times 5kx^2y^3\ \ \therefore HCF = 5x^2$
8. $15x^2y^3 = 3 \times 5x^2 \times y^2 \times y$
   $6x^3y^2 = 3 \times 2x^2 \times x \times y^2\ \ \therefore HCF = 3x^2y^2$
9. $9pqr = 3^2 \times p \times q \times r, 3q^2r = 3 \times q \times q \times r, p \times q \times r$
   $L.C.M = 9pq^2r$

10. $3x^5y^3z$ and $5x^4y^6z^5$
    LCM $= 15x^5y^6z^5$
11. $8a^2bc^3 = 2^3 \times a^2 \times b \times c^3$
    $20ac^3 = 2^2 \times 5 \times a \times c^3$
    $\therefore$ LCM $= 40a^2bc^3$
12. $-5xy^2 = -5 \times x \times y \times y, -10x^2y = -2 \times 5 \times x \times x \times y, 15x^2y^2 = 3 \times 5 \times x \times x \times y \times y$
    H.C.F $= 5xy$
13. There is no common factor, then the HCF is always 1 .
14. $A = 2^4 \times 3^5 \times x^7 \times y^9 = 2^4 \times 3^3 \times 3^2 \times x^6 \times x \times y^8 \times y$
    $B = 2^5 \times 3^3 \times x^6 \times y^8 = 2^4 \times 2 \times 3^3 \times x^6 \times y^8$
    GCD $= 2^4 \times 3^3x^6y^8$
15. $A = 8a^2\ m = 2 \times 2 \times 2a \times a \times m$
    $B = 6abmn = 2 \times 3 \times amb \times m \times n$
    $C = 10abmn^3 = 2 \times 5 \times a \times b \times m^3 \times n^2$
    L.C.M $= 2 \times 2 \times 2 \times 3 \times 5 \times a^2 \times b \times m^3 \times n^2 = 120a^2bm^3n^2$
16. $A = 3^4x^2a^3\ b^7$ then $B = 324x^2a^5\ b^9; C = 405x^4a^3\ b^7$
17. $A = 3p, B = 6p^2q \Rightarrow$ LCM $= 6p^2q$
18. $A = 15 \times x^7 \times y^8 \times z^9 \times 25x^4y^3z^3 = 3 \times 5^2 \times 5 \times x^{11} \times y^{11} \times z^{12}$
    $B = 2 \times x^6 \times y^5 \times z^3 \times 10x^2y^6z^9 = 2^3 \times 5^2 \times x^8 \times y^{11} \times z^{12}$
    G.C.D of A and $B = 25x^8 \times y^{11} \times z^{12}$
19. $P = 20x^8y^6 = 2 \times 2 \times 5 \times x^8y^6$
    $Q = 10x^4y^7z^2 = 2 \times 5 \times y^6 \times yz^2 \times x^8 \times x$
    H.C.F P and Q is $10x^8y^6$
20. $A = 2ab, B = 4bc, C = 6abc \Rightarrow$ HCF is 2 b
21. $A = a^3\ b^2$, $B = b^2c^4, C = a^3\ b^5c^4$ then H.C.F is b^2

Level – III

22) B,D	23) A,C	24) C	25) A	26) 2	27) 4
28) a -r, b-p, c-s, d-q		29) 3	30) 12	31) 4	
32) a-t, b-r, c-q, d-p		33) 3	34) 1	35) B	

22. Given $2ab = 2 \times a \times b, 2\ b^2 = 2 \times b \times b$
    LCM $= 2a\ b^2 = \dfrac{4ab^3}{2\ b}$
23. Given $14x^2y^3, 21x^2y^2, 35x^4y^5z \Rightarrow$ HCF $= 7x^2y^2$
24. HCF of ab and bc is b
    LCM of $2x^2y, 10xy$, is $10x^2y$
25. $A = 20x^2y^3$, $B = 32x^3y^2z$ and $C = 8xz^3$
    HCF of A, B is $4x^2y^2$
26. HCF of A, B, C is 4x
27. LCM of A,B,C is $160x^3y^3z^3$
28. a) LCM of $2a$ and $3b$ is $6ab$.
    b) HCF of $6a^2$ and $2a$ is $2a$.
    c) HCF of $12a^2, 24a^4$ is $12a^2$.

d) LCM of $4a^2$ and $10ab$ is $20a^2b$.

29. HCF of 6xy and 3x is $3x \Rightarrow k = 3$
30. LCM of $12a^2, 8a, 6a^2$ is $24a^2 \Rightarrow \frac{p}{2} = \frac{24}{2} = 12$
31. HCF of a^3 b^2 and -5 b^2 is b^2 (statement-I false)
    LCM of $4a^3, 6b^2$ and $8c$ is $24a^3b^2c$ (statement-II true)
32. a) The HCF of x^3y^2 and $-5y^2$ is y^2
    b) The HCF of $4x^3, 6y^2$ and $8z$ is 2
    c) The HCF of $2x^2y^3, 10x^3y^2$ and $14xy$ is $2xy$
    d) The HCF of $5x^3 - 15x^2$ and $45x$ is $5x$
33. $A = 42p^2q^2r, B = 63p^3q$ and $C = 56p^2r$
    $A = 2 \times 3 \times 7 \times p \times p \times q \times q \times r$; $B = 3 \times 3 \times 7 \times p \times p \times p \times q$
    $C = 2 \times 2 \times 2 \times 7 \times p \times p \times r$
    HCF of A and B is $21p^2q$
34. HCF of A and C is $14p^2r$
35. The LCM of numerical coefficient of A,B and C is 504.

WORKSHEET – 9

Level – I

1) B	2) B	3) C	4) D	5) C	6) D
7) A	8) A	9) A	10) C		

Level – II

1) B	2) B	3) C	4) C	5) C	6) C
7) C	8) A	9) B	10) C	11) A	12) C
13) B	14) B	15) A	16) A	17) C	18) A
19) C					

SOLUTIONS:

1. Given $a(x+5) + 2y(x+5) = (x+5)(a+2y)$
2. Given $6x(2x-y) + 7y(2x-y) = (2x-y)(6x+7y)$
3. Given $4a^2(3x-2y) + 3b^2(2y-3x) = (3x-2y)(4a^2-3b^2)$
4. Given $12x^3y^4 + 16x^2y^5 - 4x^5y^2 = 4x^2y^2(3xy^2 + 4y^3 - x^3)$
5. Given $9x^2(6x-5y) - 12x^2(6x-3y) = (6x-5y)(-3x^2) = -3x^2(6x-5y)$
6. Given $px - qy - py + qx = p(x-y) + q(x-y) \Rightarrow (x-y)(p+q)$
7. Given $a^4 + 2a^2 + 1 - a^2 = (a^2)^2 + 2a^2(1) + (1)^2 - a^2$
   $= (a^2+1)^2 - (a)^2 = (a^2+1+a)(a^2+1-a)$
8. Given $16 - a^6 + 4a^3b^3 - 4b^6 = 16 - [a^6 - 4a^3b^3 + 4b^6]$
   $= 4^2 - (a^3 - 2b^3)^2 = (4 + a^3 - 2b^3)(4 - a^3 + 2b^3)$
9. Given $axy - bcz - az + bcxy = xy(a+bc) - z(a+bc) = (a+bc)(xy-z)$
10. Given $(3p+2q)^2 - 2(3p+2q)(p+q) + (p+q)^2$ $(\because a^2 - 2ab + b^2 = (a-b)^2)$
    $= [3p+2q-(p+q)]^2 = [3p+2q-p-q)]^2 = [2p+q]^2$
11. Given $A + 36x^2y^2 + B - 26x^2y^2$
    $= 32x^7y^{12} + 24x^7y^{11} + 4x^2y^2 + 36x^2y^2 + 6x^2y^2 + 4x^5y^9 - 26x^2y^2$
    $= 32x^7y^{12} + 24x^7y^{11} + 4x^5y^9 + 20x^2y^2$
    $= 4x^2y^2[8x^5y^{10} + 6x^5y^9 + x^3y^7 + 5]$

12. Given $A = 19x^4 + 36; B = -3x^4 - 661$
$A + B = 19x^4 + 36 - 3x^4 - 661$
$= 16x^4 - 625 = (4x^2)^2 - (25)^2 = (4x^2 + 25)(4x^2 - 5^2) = (4x^2 + 25)(2x + 5)(2x - 5)$
13. Given $5px - 10qy + 2rpx - 4qry = 5(px - 2qy) + 2r(px - 2qy) = (px - 2qy)(5 + 2r)$
14. Given $3ap - 6aq - 8bq + 4bp = p(3a + 4b) - 2q(3a + 4b) = (3a + 4b)(p - 2q)$
15. Given $4xa - 6yb - 3ya + 8xb = 4x(a + 2b) - 3y(a + 2b) = (a + 2b)(4x - 3y)$
16. $pq - rs + sp - rq = q(p - r) + s(p - r) = (p - r)(q + s)$
17. $CA - BD = (5p - 3q)^2 13 \times 2^2 - (3q - 5P)13 \times 3^2 = 13(3q - 5p)[(3q - 5q)4 - 9]$
$= 13(3q - 5p)(12q - 20p - 9)$
18. Given $xy^2 + (x - 1)y = xy^2 + xy - y - 1 = xy(y + 1) - (y + 1) = (y + 1)(xy - 1)$
19. Given $x^2 - 2qx + px - 2pq = x(x - 2q) + p(x - 2q) = (x - 2q)(x + p)$

Level – III

20) B,D	21) B,D	22) C,D	23) B	24) A	25) A
26) a-s;b-p;c-q;d-r		27) A, C	28) A, C	29) A	
30) a-q;b-p;c-t;d-r		31)B, D	32) A,B	33) C	34) B 35) A

20. Given $25 - a^2 - 2ab - b^2 = 5^2 - (a + b)^2 = (5 + a + b)(5 - a - b)$
21. Given $(x + y)^2 - (x + y)(x - y) = (x + y)(x + y - x + y) = 2(x + y)y = 2xy + 2y^2$
22. Given $9p^2 - 6pq = 3p(3p - 2q) = p(9p - 6q)$
23. Given $A = 12(a + 3b)^4 - 6(a + 3b)^3 = 6(a + 3b)^3[2(a + 3b) - 1] = 6(a + 3b)^3(2a + 6b - 1)$
24. Given $B = 2a^2b(4a + 5b)^2 - 4a^3(4a + 5b) = 2a^2(4a + 5b)[b(4a + 5) - 2a]$
25. Given A+3B
$= 6(a + 3b)^3(2a + 6b - 1) + 3\{2a^2(4a + 5b)[b(4a + 5b) - 2a]\}$
$= 6\{(a + 3b)^3(2a + 6b - 1) + a^2(4a + 5b)[b(4a + 5b) - 2a]\}$
26. a)Given $ax - by - ay + bx = (a + b)x - y(a + b) = (a + b)(x - y)$
b) Given $bxy - ac + axy - bc = xy(b + a) - c(a + b) = (a + b)(xy - c)$
c) Given $bx + ba - xy - ay = x(b - y) + a(b - y) = x(b - y) + a(b - y) = (b - y)(x + a)$
d) Given $bx - bay + b^2y - ax = b(x + by) - a(x + by) = (x + by)(b - a)$
27. Given $\frac{6}{9}x^8 + \frac{4}{3}x^6 + \frac{1}{6}x^3 = \frac{12x^8 + 24x^6 + 3x^3}{18} = \frac{3x^3}{18}[4x^5 + 8x^3 + 1]$
28. Factors of $a^2 + 4a + 5a + 20 = a(a + 4) + 5(a + 4) = (a + 4)(a + 5)$
29. Statement-I: $(n^2 + 2n + 2)(n^2 - 2n + 2)$
$= n^4 - 2n^3 + 2n^2 + 2n^3 - 4n^2 + 4n + 2n^2 - 4n + 4 = n^4 + 4$
30. a) Given $10a^2 - 40b^2 = 10(a^2 - (2b)^2) = 10(a + 2b)(a - 2b)$
b) $40a^2 - 10b^2 = 10[(2a^2)^2 - b^2] = 10(2a + b)(2a - b)$
c) $a^2 - 4b^2 = (a + 2b)(a - 2b)$
d) $10a^2 - 10b^2 = 10(a + b)(a - b)$
31. Given $4x^2 + y^2 + z^2 + 4xy + 2yz + 4xy = (2x + y + z)^2$
32. Given $A + B = (a^2 + 3a)^2 - 5(a^2 + 3a) - b(a^2 + 3a) + 5b$
$= (a^2 + 3a)[a^2 + 3a - 5] - b[a^2 + 3a - 5] = (a^2 + 3a - 5)(a^2 + 3a - b)$
33. Given $12ac - b^2 - 2bc + 6ab = 6a(2c + b) - b(b + 2c) = (b + 2c)(6a - b)$

34. Given $a^3 - a^2b^2 - ab + b^3 = a^2(a - b^2) - b(a - b^2) = (a - b^2)(a^2 - b)$
35. Given $22a^4b^7 - 176a^2b^3 + 132ab^4 = 22ab^3[a^3b^4 - 8a + 6b]$

WORKSHEET – 10

Level – I

1) C	2) B	3) A	4) C	5) A	6) A
7) B	8) B	9) D	10) B	11) B	12) D
13) C	14) D	15) C			

Level – II

1) B	2) C	3) A	4) D	5) B	6) D
7) A	8) D	9) B	10) C	11) D	12) C
13) B	14) B	15) A	16) C	17) B	18) B
19) B	20) A				

SOLUTIONS:

1. Given $\sqrt{a^{10}b^8c^4} = \sqrt{(a^5b^4c^2)^2} = a^5b^4c^2$
2. Given $\sqrt{1296y^4} = \sqrt{1296} \times \sqrt{y^4} = \sqrt{(36)^2} \times \sqrt{(y^2)^2} = 36y^2$
3. $\sqrt{4x^4y^6z^8} = \sqrt{(2x^2y^3z^4)^2} = 2x^2y^3z^4$
4. Given $\sqrt{\frac{81b^2a^4}{36x^2y^6}} = \frac{\sqrt{81b^2a^4}}{\sqrt{36x^2y^6}} = \frac{9ba^2}{6xy^3} = \frac{3ba^2}{2xy^3}$
5. Given $\sqrt{x^2 + 12xy + 36y^2} = \sqrt{(x)^2 + 2(x)(6y) + (6y)^2} = \sqrt{(x + 6y)^2} = x + 6y$
6. Given $\sqrt{x^2 + 6x + 9} = \sqrt{(x)^2 + 2(x)3 + (3)^2} = \sqrt{(x + 3)^2} = x + 3$
7. Given $\sqrt{a^2 + 10a + 25} = \sqrt{(a)^2 + 2(a)5 + (5)^2} = \sqrt{(a + 5)^2} = a + 5$
8. Given $\sqrt{a^2 - 2ab + b^2} = \sqrt{(a - b)^2} = a - b$
9. Given $\sqrt{9a^2 - 6a + 1} = \sqrt{(3a)^2 - 2(3a)(1) + (1)^2} = \sqrt{(3a - 1)^2} = 3a - 1$
10. Given $\sqrt{a^{m^2} \cdot b^{n^2}} = \left(a^{m^2}\right)^{\frac{1}{2}} \times \left(b^{n^2}\right)^{\frac{1}{2}} = a^{\frac{m^2}{2}} \times b^{\frac{n^2}{2}}$
11. Given $\sqrt{4x^2 + 12xy + 9y^2} = \sqrt{(2x)^2 + 2(2x)(3y) + (3y)^2} = \sqrt{(2x + y)^2} = 2x + 3y$
12. Given $\sqrt{a^4 - 2a^2b^2 + b^4} = \sqrt{(a^2)^2 - 2a^2b^2 + (b^2)^2} = \sqrt{(a^2 - b^2)^2} = a^2 - b^2$
13. Given $\sqrt{a^4 + b^4 + c^4 + 2a^2b^2 - 2b^2c^2 - 2c^2a^2} = \sqrt{(a^2 + b^2 - c^2)^2} = a^2 + b^2 - c^2$
14. $\sqrt{36x^2 + 60xy + 25y^2} = \sqrt{(6x)^2 + 2(6x)(5y) + (5y)^2} = \sqrt{(6x + 5y)^2} = 6x + 5y$
15. Given $\sqrt{9y^2 + 2 + \frac{1}{9y^2}} = \sqrt{(3y)^2 + 2(3y)\left(\frac{1}{3y}\right) + \left(\frac{1}{3y}\right)^2} = \sqrt{\left(3y + \frac{1}{3y}\right)^2} = 3y + \frac{1}{3y}$
16. Given $\sqrt{\frac{a}{b} + \frac{b}{a} + 2} = \sqrt{\left(\sqrt{\frac{a}{b}}\right)^2 + \left(\sqrt{\frac{b}{a}}\right)^2 + 2\sqrt{\frac{a}{b} \times \frac{b}{a}}} = \sqrt{\left(\sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}\right)^2} = \sqrt{\frac{a}{b}} + \sqrt{\frac{b}{a}}$
17. Given $\sqrt{a^2 - 2ax + x^2 + 2a - 2x + 1} = \sqrt{(a - x + 1)^2} = (a - x + 1)$
18. Given $\sqrt{\frac{9p^2}{25} - 2 + \frac{25}{9p^2}} = \sqrt{\left(\frac{3p}{5}\right)^2 - 2\left(\frac{3p}{5} \times \frac{5}{3p}\right) + \left(\frac{5}{3p}\right)^2} = \sqrt{\left(\frac{3p}{5} - \frac{5}{3p}\right)^2} = \frac{3p}{5} - \frac{5}{3p}$
19. Given $\sqrt{a + b - 2\sqrt{ab}} = \sqrt{(\sqrt{a})^2 + (\sqrt{b})^2 - 2\sqrt{ab}} = \sqrt{(\sqrt{a} - \sqrt{b})^2} = \sqrt{a} - \sqrt{b}$
20. Given $\sqrt{a^2x^2 - 2ayx^2 + x^2y^2} = \sqrt{x^2(a^2 - 2ay + y^2)} = \sqrt{x^2}\sqrt{(a - y)^2} = x(a - y)$

Level – III

21) A,C 22) A 23) B 24) D 25) A 26) a-q;b-r;c-p;d-t
27) 25 28) B,D 29) D 30) a-s;b-r,t;c-p;d-q 31) A,C 32) C
33) B 34) A

21. $\sqrt{9x^{16}} = \sqrt{9} \times \sqrt{x^{16}} = 3x^8 = \frac{9}{3}x^8$
22. Given statement I : $\sqrt{(2a+3b)^2 - 24ab} = \sqrt{(2a-3b)^2} = (2a - 3\ b)$
23. Given $\sqrt{4x^2+4x+1} = \sqrt{(2x+1)^2} = 2x + 1$
24. Given $\sqrt{4a^8b^4} = \sqrt{(2)^2(a^4)^2(b^2)^2} = \sqrt{(2a^4b^2)^2} = 2a^4b^2$
25. Given $\sqrt{\frac{27a^3}{48ab^4}} = \sqrt{\frac{3\times9\times a^{3-1}}{3\times16\times b^4}} = \sqrt{\frac{9a^2}{16b^4}} = \frac{3a}{4b^2}$
26. a) Given $\sqrt{36a^2} = 6a$ b) Given $\sqrt{49x^2y^4} = 7xy^2$
    c) $\sqrt{a^8b^4} = a^4b^2$
    d) $\sqrt{100y^2} = 10y$
27. Given $\sqrt{625a^4} = 25a^2$
28. Given $\sqrt{y^2 + \frac{1}{y^2} + 2} = \sqrt{(y)^2 + \left(\frac{1}{y}\right)^2 + 2y \times \frac{1}{y}} = \sqrt{\left(y + \frac{1}{y}\right)^2} = y + \frac{1}{y}$ or $\frac{y^2+1}{y}$
29. Statement I: $\sqrt{a^{2n+2}b^{2n+4}} = \sqrt{(a^{n+1} \cdot b^{n+2})^2} = a^{n+1}b^{n+2}$
    Statement II: $\sqrt{a^{2m} \times b^{2n}} = \sqrt{(a^m b^n)^2} = a^m b^n$
30. a)Given $\sqrt{a^2+b^2-2ab} = \sqrt{(a-b)^2} = a - b$
    b) Given $\sqrt{a^2+b^2+2ab} = \sqrt{(a+b)^2} = a + b$
    c) $\sqrt{a^4+b^4+c^4+2a^2b^2-2b^2c^2-2c^2a^2} = \sqrt{(a^2+b^2-c^2)^2} = a^2+b^2-c^2$
    d) $\sqrt{a^4+b^4+c^4-2a^2b^2-2b^2c^2+2c^2a^2} = \sqrt{(a^2-b^2+c^2)^2} = a^2-b^2+c^2$
31. $\sqrt{25x^2 + \frac{5x}{2} + \frac{1}{16}} = \sqrt{(5x)^2 + 2(5x)\frac{1}{4} + \left(\frac{1}{4}\right)^2} = \sqrt{\left(5x + \frac{1}{4}\right)^2} = 5x + \frac{1}{4} = \frac{20x+1}{4}$
32. Given $f(x) = (x+2)(x+4)(x+6)(x+8) + k$
    $\Rightarrow (x + 2)(x + 8)(x + 4)(x + 6) + k \Rightarrow (x^2 + 10x + 16)(x^2 + 10x + 24) + k$
    Let $x^2 + 10x + 16 = a$ $\therefore (x^2 + 10x + 16)(x^2 + 10x + 16 + 8) + k$
    $a(a + 8) + k$
    $a^2 + 8a + k$
    $\therefore (a + 4)^2$ when $k = 16$
    $\Rightarrow (x^2 + 10x + 20)^2$
33. $\sqrt{(x+2) \times (x+8) \times (x+4) \times (x+6) + 16} = \sqrt{(x^2 + 10x + 20)^2} = x^2 + 10x + 20$
34. If 5 is added to g(x) to make it a perfect square i.e $x^2 + 10x + 20 + 5 = x^2 + 10x + 25 \Rightarrow (x + 5)^2$

8. GEOMETRY

WORKSHEET – 1

Level – I

1) B	2) A	3) B	4) C	5) D	6) A
7) B	8) A	9) B	10) A		

Level – II

1) B	2) C	3) B	4) B	5) A	6) A
7) C	8) C	9) B	10) D	11) A	12) C
13) C	14) C	15) A	16) A	17) B	18) C
19) A	20) C	21) B			

SOLUTIONS

1. Conceptual.
2. Conceptual.
3. Conceptual.
4. Conceptual.
5. $AB = \frac{8}{2} = 4$
6. 1 hectameter = 100mts
7. Conceptual.
8. Conceptual.
9. Conceptual.
10. Conceptual.
11. Conceptual.
12. Conceptual.
13. Conceptual.
14. $\frac{2}{3}AB = \frac{7}{5}BC = 42$ cm

    $\frac{2}{3}AB = 42 \Rightarrow AB = 63$ cm

    $\frac{7}{2}BC = 42 \Rightarrow BC = 12$ cm
15. $\frac{5\times4}{2} = 10$
16. $\frac{3\times2}{2} = 3$
17. $V + F = E + 2 \Rightarrow V + 4 = 6 + 2 \Rightarrow V = 4$
18. Conceptual.
19. 

    $CD = 2x + 5 = 13 \Rightarrow 2x = 4 \Rightarrow x = 4$

    $AD = 40$ cm, $AB = 14$ cm $AB + CD + AD = 14 + 13 + 50 = 67$ cm
20. $CD = 2x + 4 = 12 \Rightarrow x = 4$

    $AB = x + 4 = 8,\ AB + CD = 8 + 12 = 20$ cm

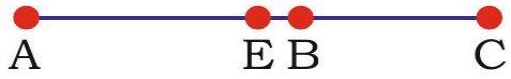
21. $AB = 32\text{ cm}, CD = 4.5\text{ cm}$
$9AB^2 - 4CD^2 = 9(32)^2 - 4(4.5)^2 = 92.61 - 81 = 11.16$

Level – III

22) A,B,C,D 23) B 24) A 25) A 26) a-r, b-s, c-q, d-p
27) B,C 28) A,C,D 29) a-s; b-r; c-q; d-p,t 30) B,C 31) C,D
32) A 33) B 34) D 35) a-r; b-s; c-q; d-p

22. Conceptual.
23. Conceptual.
24. Conceptual.
25. Conceptual.
26. Conceptual.
27. Conceptual.
28. Conceptual.
29. Conceptual.
30. Conceptual.
31. $AB = 6.5\text{ cm} \& CD = 3.5\text{ cm}$
$AB - 3CD = 13 - 10.5 = 2.5\text{ cm}$

32-34.



$\frac{3}{4}AB = 60 \Rightarrow AB = 80\text{ cm}, \frac{5}{2}BC = 60 \Rightarrow BC = 24\text{ cm} \therefore AC = 104\text{ cm}, EC = \frac{104}{2} = 52$

35. Conceptual.

WORKSHEET – 2

Level – I

1) C 2) B 3) D 4) C 5) B 6) A
7) B 8) C 9) A 10) D 11) C 12) D

Level – II

1) A 2) C 3) A 4) C 5) A 6) B
7) B 8) A 9) C 10) A 11) A 12) B
13) B 14) A 15) B 16) C 17) B 18) B
19) B 20) A

SOLUTIONS

1. $\theta = 90^\circ - \theta \Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$
2. $\theta: 180 - \theta = 4:5 \Rightarrow 5\theta = 4(180 - \theta) \Rightarrow 9\theta = 720^\circ \Rightarrow \theta = 80^\circ$
3. $50^\circ + x + 70^\circ = 180^\circ \Rightarrow x = 60^\circ$ 4. $\angle AOC = 180 - 65 = 115^\circ$
4. $(2x - 10^\circ) + (x - 5^\circ) = 90^\circ \Rightarrow 3x - 15 = 90^\circ \Rightarrow 3x = 105^\circ \Rightarrow x = 35^\circ$
5. $2x + 70 = 180^\circ \Rightarrow 2x = 110^\circ \Rightarrow x = 55^\circ$
6. $\angle POR + \angle ROQ = 180^\circ$
$\Rightarrow 5x + 7x = 180^\circ \Rightarrow 12x = 180^\circ \Rightarrow x = 15^\circ$
$\angle POR = 75^\circ, \angle ROQ = 105^\circ$

$\angle SOQ = \angle POR = 75^\circ$

8. $3x + 20 = 65 \Rightarrow 3x = 45 \Rightarrow x = 15^\circ$

9. $x : y = 4 : 11$
$x + y = 180^\circ = 4k + 11k = 180^\circ$
$\Rightarrow 15k = 180^\circ \Rightarrow k = 12$
$x = 48^\circ = a, y = 132^\circ = b$

10. $\theta - (90^\circ - \theta) = 40^\circ \Rightarrow 2\theta = 130^\circ \Rightarrow \theta = 65^\circ$

11. $x + x + 20 + x + 10 = 180^\circ$
$\Rightarrow 3x = 150^\circ \Rightarrow x = 150^\circ$

12. $3x + 3x + x + 150^\circ = 360^\circ \Rightarrow 7x = 210^\circ \Rightarrow x = 30^\circ$
$\angle BOD = x = 30^\circ$

13.
$89^\circ 60^1$
$36^\circ 24^1$
$53^\circ 36^1$

14.
$179^\circ 59^1 60^{11}$
$48^\circ 36^1 54^{11}$
$131^\circ 23^1 06^{11}$

15. Let an angle $= \theta^\circ$
$90^\circ - \theta = 180^\circ - 3\theta^\circ \Rightarrow 2\theta = 90^\circ \Rightarrow \theta = 45^\circ$

16. $2y + 5 = 75^\circ = 180^\circ \Rightarrow 2y = 100^\circ \Rightarrow y = 50^\circ$

17. $a + b = 90^\circ \Rightarrow 2x + 3x = 90^\circ \Rightarrow 5x = 90^\circ \Rightarrow x = 18^\circ \therefore a = 36^\circ, b = 54^\circ$
$b + c = 180^\circ \Rightarrow c = 180^\circ - 54^\circ = 126^\circ$

18. $x^\circ + (2x - 21)^\circ = 180^\circ$
$\Rightarrow 3x = 201 \Rightarrow x = 67^\circ$

19. $x = 2k, y = 3k, z = 5k$
$2x + 2y + 2z = 360^\circ \Rightarrow 4k + 6k + 10k = 360^\circ$
$\Rightarrow 20k = 360^\circ \Rightarrow k = 18 \therefore x = 36^\circ, y = 54^\circ, z = 90^\circ$
$a - 2b = 30^\circ$
$a + b = 180^\circ$

20.
$3b = 150^\circ$
$b = 50^\circ \therefore a = 130^\circ$

Level – III

21) A,B,D 22) A 23) A 24) B 25) C
26) a-q, b-q, c-p,s, d-p,s 27) B,D 28) a-r, b-s, c-p, d-q 29) B
30) C 31) A 32) A,B,C 33) C 34) B 35) C

21. Conceptual.
22. Statement - I :
$\theta = 2(180^\circ - \theta) - 30^\circ \Rightarrow 3\theta = 330^\circ \Rightarrow \theta = 110^\circ$
23. $x = 35^\circ$

24. $y = 140$
25. $z = 210$
26. Conceptual
27. $\angle ABC = 60°, \angle PQR = 90°$
29-31. $\angle AOB + \angle BOC + \angle COD = 180°$
$100° + 3x + 5x = 180° \Rightarrow 8x = 80° \Rightarrow x = 10°$
$\angle BOC = 30°, \angle COD = 50°, \angle AOC = \angle AOB + \angle BOC = 100° + 30° = 130°$
32. $57° + 38° = 95°$
33. $4a = 160 \Rightarrow a = 40, 5b = 150 \Rightarrow b = 30°$
34. $4x + 70 + 2y + 50 = 180 \Rightarrow 4x + 2y = 180 - 120 \Rightarrow 4x + 2y = 60 \Rightarrow 2x + y = 30$
35. $x = 25°, y = 75, z = 100$
$x + y + z = 200$

WORKSHEET – 3

Level – I

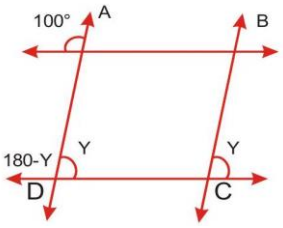
1) C	2) A	3) A	4) A	5) D	6) A
7) C	8) B	9) A	10) B		

Level – II

1) B	2) A	3) C	4) B	5) A	6) D
7) A	8) C	9) B	10) B	11) C	12) B
13) C	14) C	15) C	16) D	17) C	18) C
19) B	20) A	21) C	22) D	23) B	24) A, D

SOLUTIONS

1. Corresponding angles are equal
2. $x + 50 = 180 \Rightarrow x = 130$
3. Alternate angles are equal
4. $x = 100$
5. Opposite angles are equal
6. $Y + 35 = 85 \Rightarrow Y = 50\ X + 35 = 180 \Rightarrow X = 145\ X + Y = 195°$
7. $2x + 15 = 3x - 20 \Rightarrow x = 35$
8. $180 - 3x = 2x \Rightarrow 5x = 180 \Rightarrow x = 36$
9. $180 - y = 100 \Rightarrow y = 80$
(corresponding angles)



10. $3x - 20 = 2x + 10 \Rightarrow x = 30$
11. $2x + 30 = x + 50 \Rightarrow x = 20$ (Opposite angles are equal) $\therefore \angle 4 = 70$
$\angle 4 + \angle 5 = 180\ 70 + \angle 5 = 180 \Rightarrow \angle 5 = 110°$
12. $\angle 1 + \angle 2 = 180$
$11x + 7x = 180 \Rightarrow 18x = 180 \Rightarrow x = 10$

$\therefore \angle 1 = 110, \angle 2 = 70$

$\therefore \angle 1 = \angle 5$ [corresponding angles are equal].

13. $\angle 5 + \angle 6 = 180$

$2x + 7x = 180 \Rightarrow 9x = 180 \Rightarrow x = 20$

$\therefore \angle 5 = 40, \angle 6 = 140$

$\therefore \angle 6 = \angle 8$ [opposite angles are equal.]

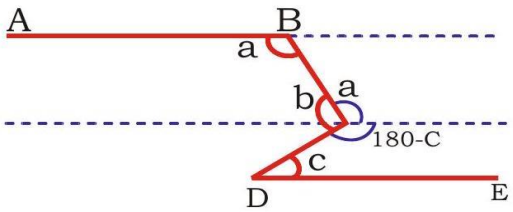
14. $\angle c = 3x$ [alternate angles are equal]

$\triangle ABC, \angle A + \angle B + \angle C = 180$

$4x + 75 + 3x = 180 \Rightarrow 7x = 180 - 75 \Rightarrow 7x = 105 \Rightarrow x = 15 \therefore \angle C = 45$

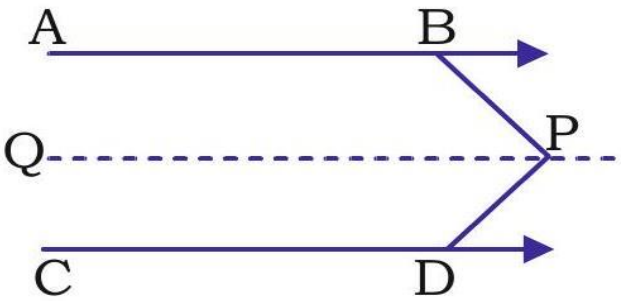
$\Rightarrow y = 180 - 45 = 135$ [linear pair]

15.



$a + b + 180 - c = 360\ a + b - c = 360 - 180 = 180$

16.

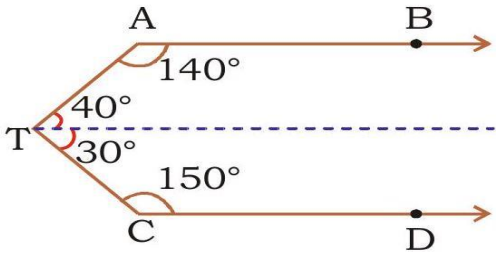


$\angle ABP + \angle BPQ = 180$

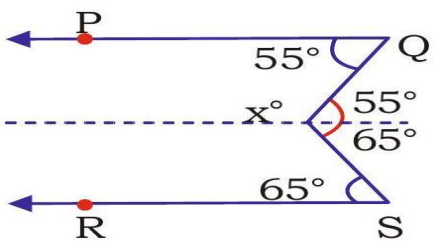
$\angle CDP + \angle QPD = 180 \therefore \angle ABP + \angle BPQ = \angle QPD = \angle CPD = 360°$

$\therefore \angle ABP + \angle BPD + \angle CPD = 360°$

17.

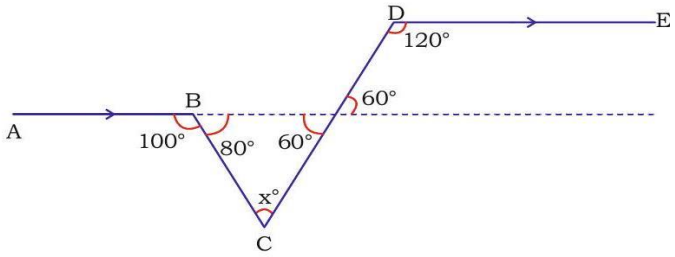


18.



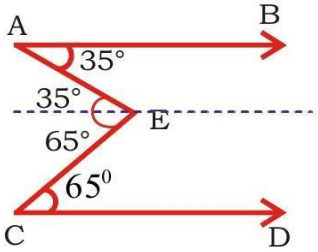
$x° + 55° + 65° = 360 \Rightarrow x = 360 - 120 = 240°$

19.



$x + 60 + 80 = 180 \Rightarrow x = 180 - 140 \Rightarrow x = x° + 55° + 65° = 360 \Rightarrow x = 360 - 120 = 40°$

20.



$35 + 65° = 100$

21. $\angle A + \angle B + \angle C = 180°$

$40° + \angle B = 65° = 180° \Rightarrow \angle B = 180 - 105\ \angle B = 75°$

$\angle ABC = \angle DCF$[Corresponding angles] $\therefore \angle DCF = 75° \Rightarrow x = 180 - 75 = 180°$

23. Conceptual

Level – III

25) A 26) C 27) B 28) A 29) a-p, b-q, c-p, d-t
30) A, B, C 31) a-r, b-p, c-s, d-t 32) B, C 33) B 34) A
35) C

24. Conceptual

25. $\angle 1 = \angle 3$ [opposite angles]

$\angle 3 + \angle 6 = 180$(sum of interior angles] $70 + \angle 6 = 180 \Rightarrow \angle 6 = 110°$

(26 – 28) $6x = 60 \Rightarrow x = 10$ [opposite angles]

$6x + z = 180$ [sum of interiror angles]

$60 + z = 180 \Rightarrow z = 120°$

$z = y$ [opposite angles]

$y = 120° \therefore 6x + y + z = 60 + 120 + 100 = 300°$

29. $z = 138°$ [corresponding angles]

$z = x$ [opposite angles]

$y + 52 = 180 \Rightarrow y = 128°$ [sum of interior angles]

$x + z - y = 138 - 138 - 128 = 148°$

30. $q = 64°$ [alternate angles]

p = q [alternate angles]

$p + q = 128$

31. $y^\circ = 120$ [corresponding angles]
t = 120 [opposite angles]
$50 + z = 180^\circ$ [exterior angles]
$\Rightarrow z = 130^\circ$
$z + x = 180^\circ$ (linear pair) $\Rightarrow x = 50^\circ$

32. $\angle CBP = 128^\circ$
$\therefore \angle PBD = 180 - 128^\circ = 52^\circ = \angle APD \ \therefore \angle ABP = \angle CBP - \angle ABC = 128 - 90 = 32$

33-35 $x = a$ [opposite angles]
$x + y = 180^\circ$ [interior angles], $b + y + p = 360^\circ$
$\Rightarrow b + y = 360 + 105 = 195^\circ$
$y + b = 195^\circ$
$b = z$\{ A.I.A \}
from (1) & (2) $y + z = 195^\circ \Rightarrow y + \frac{2y}{3} = 195$
$\frac{5y}{3} = 195 \Rightarrow y = 39 \times 3 = 117^\circ$ As $y + b = 195$
$b + 117 = 195 \Rightarrow b - 195 - 117 = 78$
As $b + w = 180^\circ \Rightarrow 78 + w = 180$
$w = 102^\circ$
As $x + y = 180^\circ \Rightarrow x + 117 = 180 \Rightarrow x = 180 - 228 = 63^\circ$

WORKSHEET 04

Level – I

1) C	2) A	3) B	4) C	5) A	6) A

Level – II

1) D	2) C	3) D	4) A	5) B	6) C
7) A	8) B	9) A	10) C	11) A	12) B
13) B	14) D	15) D	16) D		

SOLUTIONS

1. $\frac{EG}{GI} = \frac{FH}{HJ}$, Let HJ = x cm
$\frac{2}{4} = \frac{3}{x} \Rightarrow x = \frac{12}{2} = 6$ cm

2. $\frac{PQ}{PL} = \frac{PR}{PM}$
$\frac{PL + LQ}{PL} = \frac{PR}{PM}$
$\frac{8}{3} = \frac{5.6}{PM} \Rightarrow PM = \frac{5.6 \times 3}{8} = 0.7 \times 3 = 2.1$ cm

3. $\frac{CL}{LA} = \frac{CM}{MD}, \frac{CL}{AL} = \frac{CE}{AL}, \frac{CE}{EB} = \frac{CM}{MD}$ 4. $\frac{BM}{BC} = \frac{AL}{AC} \Rightarrow \frac{x}{22} = \frac{4}{11} \Rightarrow x = 8$

4. $\frac{PQ}{PL} = \frac{PR}{PM} \Rightarrow \frac{PL+LQ}{PL} = \frac{PR}{PM} \Rightarrow \frac{5+8}{5} = \frac{3.9}{PM} \Rightarrow PM = \frac{3.9 \times 5}{13} = 0.3 \times 5 = 1.5$ cm

5. $\frac{X}{18-X} = \frac{4}{6} \Rightarrow 6X = 72 - 4X \Rightarrow 10X = 72 \Rightarrow X = 7.2$

6. $\frac{X}{18-X} = \frac{4}{6} \Rightarrow 6X = 72 - 4X \Rightarrow 10X = 72 \Rightarrow X = 7.2$
$\therefore RT = 18 - X = 18 - 7.2 = 10.8$ cm

8. $\frac{PR}{RT} = \frac{QS}{su} \Rightarrow \frac{3}{4} = \frac{Qs}{su}$

$$\frac{3}{3+4}=\frac{QS}{QS+Su}\Rightarrow\frac{3}{7}=\frac{Qs}{Qu}$$

9. $\frac{AB}{BC}=\frac{1}{2}=\frac{PQ}{QR}\Rightarrow\frac{PR}{RQ}=\frac{PQ+QR}{RQ}=\frac{1+2}{2}=\frac{3}{2}$
10. $\frac{AG}{BH}=\frac{CE}{DF}\Rightarrow\frac{120}{100}=\frac{30}{DF}=DF=25\text{ cm}$
11. $\frac{AG}{BH}=\frac{EG}{FH}\Rightarrow\frac{120}{100}=\frac{40}{FH}=FH=\frac{4000}{120}=\frac{100}{3}$ cm
12. $\frac{AB}{BC}=\frac{BD}{EC}\Rightarrow\frac{12}{16}=\frac{BD}{4}=BD=3\text{ cm}$
    $AD=AB-BD=AD=12-3=9\text{ cm}$
13. According to the converse of midpoint of theorem $AE=EC=\frac{1}{2}AC$
    Already we know that $AB=AC$ and $AD=DB=\frac{1}{2}AB$\{AsD is the midpoint \}
    from (1) & (2)
    $AE=AD$
    $\therefore\triangle ADE$ is an isosceles triangle.
14. $\frac{AD}{DE}=\frac{AB}{BC}\Rightarrow\frac{4}{7}=\frac{20}{BC}\Rightarrow BC=35\text{ cm}$
15. $\frac{x}{x-2}=\frac{x+2}{x-1}\Rightarrow x(x-1)=x^2-4\Rightarrow x^2-x=x^2-4\Rightarrow x=4$
16. $\frac{AD}{AB}=\frac{AE}{AC}=\frac{2}{AD+DB}=\frac{AE}{9}\Rightarrow\frac{2}{8}=\frac{AE}{9}\Rightarrow AE=\frac{18}{8}=2.25$

Level – III

17) A,B 18) D 19) B 20) A 21) D 22) a-4, b-1, c-2, d-3
23) A,B 24) D 25) A 26) B 27) A,B,C 28) B
29) B 30) B 31) C

17. According to converse of midpoint of theorem
    E is the midpoint of AC
    F is the midpoint of BC
18. Conceptual
19. $\frac{AD}{AB}=\frac{AE}{EC}\Rightarrow\frac{2}{4}=\frac{5}{EC}\Rightarrow EC=10\text{ cm}$
20. $\frac{AB}{AC}=\frac{DB}{EC}\Rightarrow\frac{10}{12}=\frac{DB}{8}\Rightarrow DB=\frac{8\times10}{12}=6\frac{2}{3}$
21. $\frac{AD}{DB}=\frac{AE}{EC}\Rightarrow\frac{4}{6}=\frac{3}{EC}\Rightarrow EC=\frac{18}{4}=4.5$
22. Conceptual
23. Conceptual
24. $\frac{CD}{DR}=\frac{AB}{AS}\Rightarrow\frac{30}{DR}=\frac{27}{15}\Rightarrow DR=\frac{50}{3}=16\frac{2}{3}$
25. $\frac{DC}{DR}=\frac{AB}{QS}\Rightarrow\frac{30}{PR}=\frac{27}{9}\Rightarrow PR=10\text{ cm}$
    DC-PR $=30-10=20$ cm
26. $DP+RC\Rightarrow DC-PR=30-10=20\text{ cm}$
27. $\frac{CD}{DF}=\frac{AB}{AE}\Rightarrow\frac{60}{DF}=\frac{30}{5}\Rightarrow DF=10\text{ cm}$
    $\frac{CD}{AB}=\frac{FH}{EG}\Rightarrow\frac{60}{30}=\frac{FH}{10}\Rightarrow FH=20\text{ cm}$
    $\frac{CD}{AB}=\frac{CH}{GB}\Rightarrow\frac{60}{30}=\frac{CH}{15}\Rightarrow CH=30\text{ cm}$

$\therefore DF + FH + CH = 60$ cm

28. Conceptual

29. $\frac{AD}{DB} = \frac{AE}{EC} \Rightarrow \frac{4x-3}{2x-1} = \frac{8x-7}{5x-3} \Rightarrow 20x^2 - 12x - 15x + 9 = 24x^2 - 29x + 7$
$\Rightarrow 4x^2 - 2x - 2 = 0$
$\Rightarrow 2x^2 - x - 1 = 0 \Rightarrow 2x^2 - 2x + x - 1$
$2x(x-1) + 1(x-1) = 0$
$(2x+1)(x-1) = 0$
$\Rightarrow x = 1$

30. $AB = AD + DB = 1 + 2 = 3$

31. $AC = AE + EC = 1 + 2 = 3$

WORKSHEET – 5

Level – I

1) 1	2) 4	3) 1	4) 2	5) 3	6) 3
7) 1	8) 2	9) 4	10) 2	11) 2	12) 1 13) 1

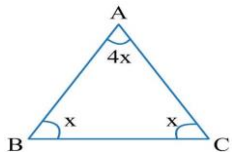
Level – II

1) 1	2) 3	3) 4	4) 1	5) 2	6) 3
7) 2	8) 3	9) 2	10) 2	11) 1	12) 3
13) 1	14) 2	15) 2	16) 1	17) 2	18) 1
19) 3	20) 2	21) 3	22) 2	23) 3	

SOLUTIONS

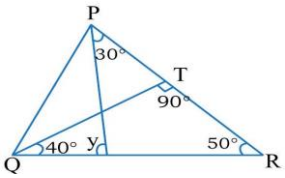
1. $\angle A + \angle B + \angle C = 180^\circ \Rightarrow \angle A = 180^\circ - 105^\circ - 50^\circ = 25^\circ$
2. The angles are $2x, 3x, 4x$
$2x + 3x + 4x = 180^\circ \Rightarrow 9x = 180^\circ \Rightarrow x = 20^\circ$ $\because$ largest angle $= 4 \times 45^\circ = 80^\circ$
3. $\angle QPR = 180^\circ - 135^\circ = 45^\circ$
$\angle QPR = 180^\circ - 110^\circ = 70^\circ$ $\therefore \angle QPR = 180^\circ - 70^\circ - 45^\circ = 65^\circ$
4. $\angle BAC = \angle EAF = 45^\circ$
$\angle ACB = 180^\circ - 105^\circ = 75^\circ$ $\therefore \angle ABC = 180^\circ - 75^\circ - 45^\circ = 60^\circ$
5. $\angle A + \angle B + \angle C = 180^\circ \Rightarrow \angle C = 40^\circ$
$\therefore$ Smallest side is opposite to the smallest angle i.e., $\overline{AB}$
6. Sum of two sides is greater than third side and difference of two sides is less than third side
7. Conceptual
8. $AB \parallel CD, AD$ is transversal $\therefore \angle CDE = \angle BAE = 52^\circ \therefore \angle CED = 180^\circ - 40 - 52^\circ = 88^\circ$
9. The three angles are $2x, 3x, 10x$
$2x + 3x + 10x = 180^\circ \Rightarrow x = \frac{180^\circ}{15} = 12$ $\therefore$ The largest angle is $10 \times 12 = 120^\circ$
10. Sum of angles in triangle $AEC = 180^\circ$
Sum of angles in a triangle $FDB = 180^\circ$ $\therefore \angle A + \angle B + \angle C + \angle D + \angle E + \angle F = 360^\circ$
11. In $\triangle ABC, \angle A = 80^\circ, \angle BCO = 30^\circ,$
$\therefore \angle B = 180^\circ - 80^\circ - 60^\circ = 40^\circ \Rightarrow \angle OBC = 20^\circ \Rightarrow \angle BOC = 180^\circ - 30^\circ = 130^\circ$
12. $\angle ABC = 180^\circ - \angle EBA = 60^\circ, \angle ACB = 180^\circ - \angle ACD = 70^\circ$
$\therefore \angle BAC = 180^\circ - 60^\circ - 70^\circ = 50^\circ$

13.



$\Rightarrow 6x = 180° \Rightarrow x = 30°$

14.



15. Angle are $2x, 3x, 4x$
$9x = 180° \Rightarrow x = 20°$ $\therefore \angle A = 40°, \angle B = 60°, \angle C = 80°$ $\therefore$ Smallest side is BC
16. $\angle B = 180° - 120° - 30° = 30° = x°$
$y^o = x^o = 30°$($\because$ BC ‖ DE, is transveral)
$z° = 120°$ $\therefore x + y + z = 30 + 30 + 120° = 180°$
17. $\triangle$ ABC is isosceless triangle
$\angle BAC = 180° - 110° = 70°$ $\therefore \angle ABC = \angle ACB = 55°$
18. Let $\angle A$ be the smallest angle
$\angle B = 2\angle A$ and $\angle C = 3\angle A$
$\angle A + \angle B + \angle C = 180° \Rightarrow \angle A + 2\angle A + 3\angle A = 180°$
$6\angle A = 180° \Rightarrow \angle A = 30°$
19. Verify
Sum of two sides of atriangle is greater than third side difference of two side of a triangle is less than third side
20. $\angle A + \angle B + \angle C + \angle D = 360° \Rightarrow \angle D = 360° - 130° = 230°$ $\therefore X = 360° - 230° = 130°$
21. $\triangle$ ABM, $\angle BAM = 25°$
$\angle A = 180° - 65° = 33° = 82° \Rightarrow \angle BAN = 41°$
($\therefore$ Anis besictor of $\angle A$) Now $\angle MAN = \angle BAN - \angle BAM = 41° - 25° = 16°$
22. $\angle 1 + \angle 2 + \angle 3 = 180°$ $\angle 1 = 180° - 110° = 70, \angle 3 = 180° - 140° = 40$

Level – III

24) A 25) C 26) C 27) a-r; b-p; c-s;d-p,q 28) C
29) B 30) A 31) A, B, C, D 32) B 33) C
34) D 35) B, D

23. $\angle X + \angle Y + \angle Z = 180° \Rightarrow \angle Z = 180° - 140° = 40°$
$\therefore$ Smallest side is opposite to the smallest angle $\therefore$ xy is smallest side (24 to26)
$\angle A + \angle B + \angle C = 180° \Rightarrow 4X - 20 = 180° \Rightarrow X = 50°$ $\therefore \angle A = 90°, \angle B = 40°,$
$\angle C = 50°$
$\therefore \angle A$ is right angle
$\angle B, \angle C$ are acute angles
$\therefore \triangle$ ABC is right angled triangle
27. $\angle A + \angle B + \angle C = 180° \Rightarrow \angle C = 180° - 112° - 24° = 44°$
a) $\angle C = X = 44°$
b) $\angle A + \angle C = 112° + 44° = 156°$

c) $\angle C + \angle B = 44° - 24° = 20°$

d) $2\angle A - \angle B + \angle C) = 2(112°) - (44° + 24°) = 224° - 68° = 156°$

(28-30) $A = B + 15°$

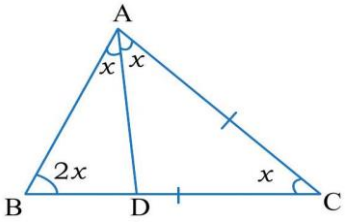
$C = B - 30°$

$A + B + C = 180°\ B + 15 + B + B - 30° = 180°$

$3\ B = 19 \Rightarrow 5\ B = 65°$

31. Conceptual.

(32 – 34)



$\Rightarrow 5x = 180° \Rightarrow x = 36°$

35. $AB = AC \Rightarrow 3y = 7x$

$AD = AE \Rightarrow 9x = 13 + 2y$

$\Rightarrow 9\left(\frac{3y}{7}\right) = 13 + 2y \Rightarrow \frac{27}{7}y - 2y = 13 \Rightarrow \frac{13}{7}y = 13$

$\Rightarrow y = 7 \ \therefore x = 3$

WORKSHEET 06

Level – I

1) A	2) D	3) B	4) D	5) C	6) B

Level – II

1) A	2) A	3) A	4) A	5) A	6) A
7) B	8) A	9) C	10) A	11) A	12) B
13) B	14) A	15) 3	16) A	17) B	18) B
19) C	20) A	21) C	22) 2		

SOLUTIONS

1. $AB = CF$

$AB + BF = CF + BF$

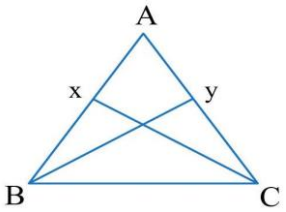
$AF = BC$ {given }

$EF = BD$ {given }

$\lfloor AFE = \lfloor CBD$ {given }

By SAS congruency $\triangle AFE \cong \Delta CBD$

2.



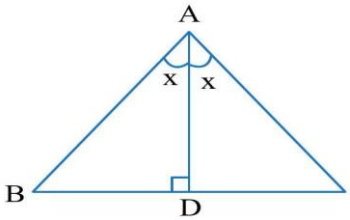
AB = AC and AX = AY
$\Rightarrow \angle ABC = \angle ACB$ and $BX = CY$
$\Delta \times BC \cong \triangle \times CB$ by SAS
$\Rightarrow$ CX = BY
$\therefore \triangle AXC \cong \triangle AYB$ by S.S.S axiom

3. AC = AD {given }
AB = AB{ common }
$\underline{CAB} = \underline{DAB}$ {AB bisects $\angle A$}
ΔCAB $\cong \triangle$ DAB

4. BCD = $\underline{ADC}$ { given
$\underline{ACB} = \underline{BDA}$ {given
from (1) & (2)
$\underline{BCD} + \underline{ACD} = \underline{ADC} + \underline{BDA}$
$\therefore \angle ACD = \angle BDC$ and $\angle ADC = \angle BCD$
and CD is common
By ASA congruency
$\triangle$ ACD $\cong \triangle$ BDC

5. As $l \parallel m$ and $p \parallel q$AC acts as transversal $\angle 3 = \angle 2, \angle 1 = \angle 4$ { alternate interior angles}
AC is common by ASA congruency
$\triangle BAC \cong \triangle DCA$

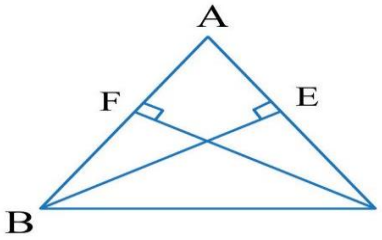
6.



$\triangle ADB \cong \triangle ADC$ by A.S.A
$\therefore$ AB = AC(By C.P.C.T.C)

7. Given that $AB = AC \lfloor ABC = \lfloor ACB$ {Base angles of an Isosceles Δ}
$\underline{EBC} = \underline{FCB}$ {BE and CF are angle bisector of $\angle B \& \angle C$
BC is common By ASA Congruency
$\triangle EBC \cong \triangle FCB$

8.



Given BE = CF
$\triangle CAF \cong \triangle BAE$ by A.A.S.
$\therefore$ AC = AB (by C.P.C.T.C)

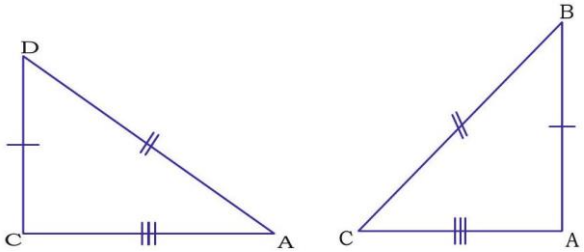
9. $\angle BAD = \angle BCE$ (Given)
AB = AC (Given)

$\angle ABD = \angle CBE$ (Common angle)
$\therefore \triangle ABD \cong \triangle CBE$ by A.S.A

10. $AB = EF$ (Given)
$\lfloor ABD = \angle FEC = 90°$
$BC = DE$ { given }
$BC + CD = BE + CD$
BD = CE
By SAS Congruency
$\triangle ABD \cong \triangle FEC$

11. PS = PR
$\Rightarrow \angle PRS = \angle PSR \Rightarrow 180 - \angle PRS = 180 - \angle PSR$
$\Rightarrow \angle PRT = \angle PRQ \therefore \triangle PST \cong \triangle PRQ$ by A.S.A

12. $AB = CD$ {given}



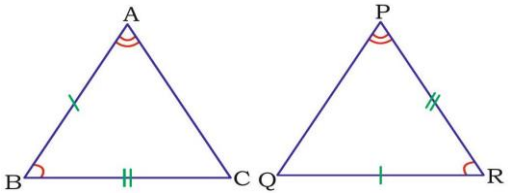
AD = BC
AC = AC{common}
By SSS congruency
$\triangle CDA \cong \triangle ABC$

13. $\angle DBC = \angle DCB \Rightarrow$ CD = BD
and AB = AC $\Rightarrow \angle ABC = \angle ACB$
$\therefore \triangle ABD \cong \angle AZD$ by S.A.S

14. $LM = MN$ {given}
QM = MR
$\underline{QLM} = \underline{MNR} = 90°$ {given}
By RHS Congruency
$\triangle MLQ \cong \triangle MNR$

15. $\lfloor DAB = \lfloor CBA = 90°$
Ay = Bx (given)
AB = BA (common)
By RHS congruency
$\triangle BAX \cong \triangle ABY$
$\triangle BAY \cong \triangle ABX$

16.



By SAS congruency

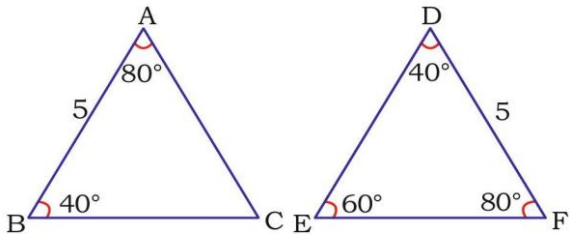
17. Two angles and an included side
∴ ASA Congruency

18. $\triangle ABC \cong \triangle PQR$
$\angle A = \angle P$ {CPCT}
$3X = 2X + 20\ X = 20°$
$\angle P = 2x + 20 = 2(20) + 20 = 60°$

19. $\triangle A \times B \cong \triangle AZD$ (By R.H.S)
$BX = DZ = DY + YZ = 3 + 2 = 5$
$BY = BX + XY = 5 + 2 = 7$

20. $ASAB = AC \Rightarrow \angle B = \angle C = 35°$
$\triangle ABC$ is and isosceles triangle In isosceles triangle, medium divides the base as well as the angle at vertex into two equal halves.
$\angle A + \angle B + \angle C = 180°$ $\angle A + 35 + 35 = 180°$
$\angle A = 110°$ $\angle BAD = \frac{1}{2}\angle A = \frac{110}{2} = 55°$

21.



As $\triangle ABC \simeq \triangle FDE$
$\underline{A} = \lfloor F = 80°$
$AB = FD = 5$ cm
$\underline{B} = \lfloor D = 40°$
$\underline{C} = \underline{E} = 60°$

22. let $BAC = x$
$\underline{PAC} = \underline{PAB} + \underline{BAC} = 90 + x$
$\underline{BAR} = \underline{RAC} + \angle BAC = 90 + x$
In $_{\Delta}PAC$ and $_{\triangle}BAR$
$PA = AB$
$AC = AR$ { Sides of a square }
$\underline{PAC} = \underline{BAR}$ { given }
By SAS congruency
$\triangle PAC \cong \triangle BAR$

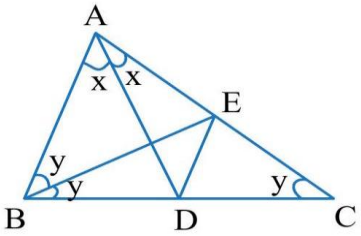
Level – III

23) A,B,C,D 24) B 25) A 26) A 27) a-r; b-s,t ; c-p,q; d-p,q:
28) A,B,C,D 29) B 30) B 31) B 32) a-q, b-q, c-r, d-q

23. As$\underline{PQT} = \underline{RQU}$ {given } → (1)
$\underline{TQS} = \underline{VQS}$ { given } → (2)
Adding (1) & (2)
$\underline{PQS} = \underline{RQS}$ → option(1)

Given that PQ = QR and QS common side and
$\underline{PQS} = \underline{RQS}$ By SAS congruency
ΔPQS ≅ RQS → option (2) ∴ QPS = $\underline{\text{QRS}}${CPCT} → option(3)
As $PQ = RQ$ {given }
| PQT = ∠RQU {given }
$\underline{\text{QPT}} = \underline{\text{QRU}}$ { proved }
By ASA congruency
ΔQPT ≅ QRU → option (4)

24. $\underline{\text{P}} = \underline{\text{Q}} = \underline{\text{R}} = \underline{\text{S}} = 90°$ {interior angle of a square }
⌊T = ⌊TSR = ⌊TRS = 60° {interior angle of a equilateral triangle }
⌊TSP = ⌊TSR + ⌊PSR = 90 + 60 = 150°
25. As TS = SP [Equilateral triangle is described on squares ∴ length sides of squares and length of sides of triangle are equal }
$\underline{\text{TPS}}$ = ⌊PTS 150 + ⌊TPS + ⌊TPS = 180° ⇒ ⌊TPS = 15°
26. As TS = TR
SP = RQ
$\underline{\text{TSP}}$ = ⌊TRQ = 150°
By SAS congruency
ΔTSP ≅ ΔTRQ
27. $\triangle ABC \cong \triangle PQR$
$\angle A = \angle P = 70° \ \angle B = \angle Q = 50°$
$\angle C = \angle R = 60°$
28. In △ ACE and △ BCD
$\underline{\text{EAC}} = \underline{\text{DBC}}$ { given } → (1)
$\underline{\text{DCA}} = \underline{\text{ECB}}$ {given } Add ∠DCE on both sides
$\underline{\text{DCA}} + \underline{\text{DCE}} = \underline{\text{ECB}} + \underline{\text{DCE}}$ |ECA = |DCB → (2) (option) → 4
AC = BC{ given } → (3)
By SAS congruency rule ΔDBC ≅ ΔEAC → option (1)
DB = EA DC = EC{CPCT} → option (2,3)
29. AB = BC = CD = DA{ sides of square }
AP = BQ = CR {given }
CQ = BC − BQ CQ = AB − AP CQ = BP ∠C = $\underline{\text{B}}$ = 90°
By △ SAS congruency
△ QCR ≅△ PBQ
QR = QP{CPCT}
$\underline{PRQ} = \underline{PQR}$ = 45°{ angle opp to equal sides }
(30,31). QPB = 45°, RQC = 45°
32.



Construction : BE is angle bisector of $\angle ABC$ Join DE

Proof: in $\triangle BEC, \angle EBC = \angle ECB$

$\Rightarrow CE = BE$

$\triangle ABE \cong DCE$ by S.A.S

AB = CD (Given)

$\angle ABE = y = \angle DCE$ (by construction)

BE = CE

$\therefore$ AE = ED (by C.P.C.T.C) and $\angle BAE = \angle CDE = 2x$

From $\triangle AED, = AE = ED$

$\Rightarrow \angle AED, = AE = ED \Rightarrow \angle ADE = \angle DAE = x$

From $\triangle ABD$,

$\angle ADC = \angle DBA + \angle DAB$ (exteriror angle property)

$\Rightarrow \angle ADE = \angle EDC = \angle DBA + \angle DAB$

$\Rightarrow x + 2x = 2y + x$

$\Rightarrow x = y$

From $\triangle ABC, \angle A + \angle B + \angle C = 180^\circ = 2x + 2y + y = 180^\circ$

$= 5x = 180^\circ \Rightarrow x = 36^\circ = y$

$\angle BAC = 2x = 72^\circ$

$\angle ABC = 2Y = 72^\circ$

$\angle ADC = 3x = 108^\circ$

$\angle ADB = 180^\circ - \angle ADC = 72^\circ$

WORKSHEET – 7

Level – I

1) C	2) B	3) A	4) D	5) D	6) C
7) B	8) D	9) B			

Level – II

1) A	2) D	3) B	4) A	5) C	6) A
7) A	8) B	9) A	10) D	11) C	12) C
13) C	14) C	15) B			

1. $\angle A + \angle B + \angle C + \angle D = 360^\circ \Rightarrow \angle D = 360^\circ - (\angle A + \angle B + \angle C) = 360^\circ - 220^\circ = 140^\circ$
2. $\angle A + \angle B + \angle C + \angle D = 360^\circ \Rightarrow 12X + 36^\circ = 360^\circ \Rightarrow 12X = 324 \Rightarrow X = 27$
3. $\angle P + \angle Q + \angle R + \angle S = 360^\circ \Rightarrow \angle R + \angle S = 360^\circ - 100^\circ - 80^\circ = 180^\circ$
4. $\angle R = 360^\circ - (P + \angle Q + \angle S) = 360^\circ - 290^\circ = 70^\circ$
5. $EF = \frac{1}{2}(AB + CD) = \frac{1}{2}(16) = 8CM$
6. (1) The angles of a quadrilateral are given as in the ratio 2: 3: 6: 7

   "Let the angles be 2x, 3x, 6x, 7x.

   "The sum of angles of a quadrilateral = 360°

   $\Rightarrow 2x + 3x + 6x + 7x = 360 \Rightarrow 18x = 360 \Rightarrow x = 20$

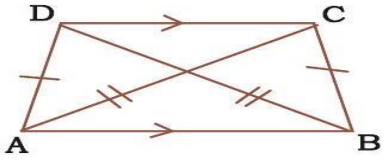
$\therefore$ The largest angle $= 7x = 7 \times 20 = 140^\circ$

7. $EF = \frac{1}{2}(AB + CD) \Rightarrow 10 = \frac{1}{2}(10 + CD)$
   $\Rightarrow CD = 10$ cm
8. $\frac{OA}{OC} = \frac{OB}{OD} \Rightarrow \frac{5}{10} = \frac{OB}{OD} \Rightarrow OD : OB = 2 : 1$
9. ABCD is an isosceles trapezium $\Rightarrow AC = BD \Rightarrow BD = 10$ cm
10. Sum of all angles in a quadrilateral is 360°
    $\therefore$ Fourth angle $= 360^\circ - (90^\circ + 65^\circ + 55^\circ) = 360^\circ - 210^\circ = 150^\circ$
11. $(\text{Diagonal})^2 = \text{side}^2 + \text{side}^2 = 5^2 + 5^2 = 5\sqrt{2}$ cm
12. (3) In a trapezium ABCD in which $\overline{AB} \parallel \overline{CD}$
    Then $LA + LD = 180^\circ$ = Straight line angle
    $\angle B + \angle C = 180^\circ$ = Straight angle
13. $AC^2 = AB^2 + BC^2$
    $AC^2 = 2AB^2 \Rightarrow 100 = 2AB^2 \Rightarrow AB = 5\sqrt{2}$ cm
14. $OA = OC \Rightarrow 3y - 4 = y + 20 \Rightarrow 2y = 24 \Rightarrow y = 12$
15. (2) $OA = OB \Rightarrow \underline{OAB} = \underline{OBA} = 26^\circ \Rightarrow \underline{COB} = \underline{OAB} + \underline{OBA} = 52^\circ$

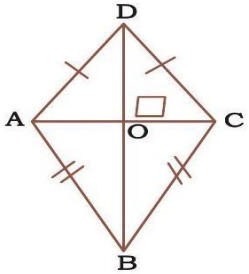
Level – III

16) A,B,C,D	17) B,C	18) B,C	19) A	20) C	21) C
22) A	23) a-r, b-q, c-p, d-s		24) 180	25) C	26) B
27) 3	28) a-q, b-r, c-s, d-p		29) A	30) C	31) B
32) A					

16. Conceptual
17. Conceptual
    (2,3)



In isosceles trapezium ABCD, $\overline{AB} \parallel \overline{CD}$, $AB = BC$
then the length of diagonals are equal



A kite ABCD with $AB = AC$ and $AD = CD$
the diagonals of a kite are perpendicular to each other i.e, $BD \perp AC$

19. Conceptual
20. Conceptual
21. Conceptual
22. Conceptual

23. Conceptual
24. $\angle A + \angle B + \angle C + \angle D == 360^\circ \Rightarrow \angle C + \angle D = 360^\circ - (120^\circ - 60^\circ) = 180^\circ$
25. Length of median $EF = \frac{1}{2}(AB + DC)$
26. $EF = \frac{1}{2}(6 + 4) = 5$ cm
27. $4 = \frac{1}{2}(3 + CD) \Rightarrow 8 = 3 + CD \Rightarrow CD = 5$ cm
28. Conceptual
29. (1) In $\triangle ABC$, we have $\angle CAB + \angle ABC = 180^\circ$. and $\angle ABC = 90^\circ \Rightarrow 32^\circ + 90^\circ + \angle ACB = 180^\circ$
    $\angle ACB = 58^\circ$
30. (3) We know that the diagonals of a rectangle are equal and bisect each other
    $\therefore OA = OB \Rightarrow \angle OBA = \angle OBA = 32^\circ$
    $\angle ABC = 90^\circ \Rightarrow \angle OBA + \angle OBC = 90^\circ$
    $32^0 + \angle OBC = 90^\circ$
    $\therefore \angle OBC = 58^\circ$
31. (2) In a rightangle triangle the square of hypotenuse is equal to sum of the squares of opposite side and adjacent side i.e. $AC^2 = AB^2 + BC^2$
32. Area of the rhombus is $\frac{1}{2}\ d_1\ d_2 = \frac{1}{2} \times 24 \times 18 = 216\ cm^2$
33. $OA = OC = \frac{AC}{2} = 12$ cm
34. $OB = OD = \frac{BD}{2} = 9CM$
35. Conceptual.

WORKSHEET – 8

Level – I

1) A	2) A	3) B	4) C	5) A	6) B	7) C
8) D	9) C	10) C				

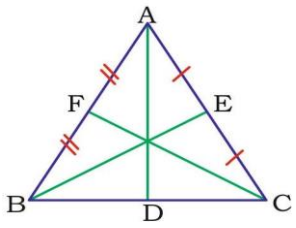
Level – II

1) A	2) B	3) C	4) D	5) A	6) D	7) C
8) C	9) D	10) A	11) C	12) A	13) A	14) A
15) A	16) A	17) B	18) C	19) B	20) A,B,C	21) B
22) C	23) D	24) 6	25)a-s,b-r,c-p,d-q		26) A,B	27) C
28) C	29) C	30) A,B,C	31) A,B	32) a-s,b-q,c-p,d-r		

SOLUTIONS

1. Conceptual.
2. Conceptual.
3. Conceptual.
4. Conceptual.
5. Conceptual.
6. Conceptual.
7. Conceptual.
8. Conceptual.

(9,10,11).



$BE = CE > BC$

(1)

$BF = CF > BC$

$BE + CE + BF + CF > 2BC$

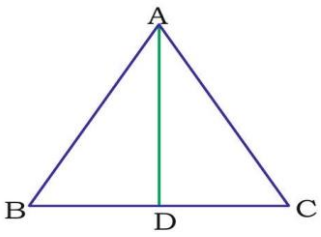
$BE + \frac{AC}{2} + \frac{AB}{2} + CF > 2BC$

$BE + CF > 2BC - \left[\frac{AB + AC}{2}\right]$

$BE + CF > 2BC - \frac{BC}{2}$

$BE + CF > \frac{3}{2}BC$

Similarly $AD + BE > \frac{3}{2}AB$ and $AD + CF > \frac{3}{2}AC$

12. $AB + AC > 2 \times AD$ ∴ on comparision $K = 2$
13. $AB + BC > 2 \times BE$ ∴ $k = 2$
14. $BC + AC > 2 \times CF$ ∴ $k = 2$
15. $BC + AC < \frac{4}{3}(AD + BE + CF)$
16. 

Angle bisector divides the base in the ratio of other two sides.

$AB : AC = BD : CD$

17. Conceptual
18. Conceptual
19. Conceptual
20. Conceptual
21. $AG = \frac{2}{3}AD$ $AG = 6 \Rightarrow AD = 9$ $GD = \frac{1}{3}AD \Rightarrow GD = 3$
22. $BE = 12$

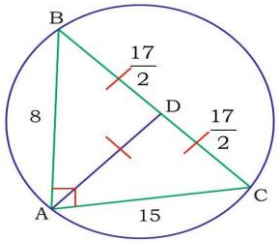
    $GE = \frac{1}{3}CF \Rightarrow GE = 4$
23. $CG = \frac{1}{3}CF \Rightarrow CF = 15$
24. $HS = 18$

    $GS = \frac{1}{3}HS \Rightarrow GS = 6$
25. Conceptual

26. Conceptual
27. $BC^2 = AB^2 + AC^2 = 64 + 225 = 289$
28. Circum centre is equidistant from vertices

$SA = SB = SC = \frac{1}{2}BC$

29.



$BC^2 = AB^2 + AC^2$

$= (8)^2 + (15)^2$

$BC^2 = (17)^2 \Rightarrow BC = 17$

Circumcentre lies exactly the midpoint of 37ypotenuse circum centre is equidistant $AD = \frac{17}{2}$ cm from the each vertex.

$AD = BD = CD = \frac{17}{2}$ cm

30. $AG = \frac{2}{3}AD\ BG = \frac{2}{3}BE\ CG = \frac{2}{3}CF$

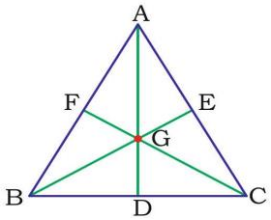
$AB^2 + BC^2 + AC^2 = 3(AG^2 + BG^2 + CG^2)$

$= 3(64 + 100 + 144) = 3 \times 308 = 924$

31. $\frac{BD}{CD} = \frac{AB}{AC} = \frac{15}{21} = \frac{5}{7}$

$\frac{AI}{ID} = \frac{AB + AC}{BC} = \frac{15 + 21}{18} = \frac{36}{18} = \frac{2}{1}$

32.



(a) $AG : GD = 2 : 1$

$\frac{AG}{AD} = \frac{2}{3} \Rightarrow \frac{6}{AD} = \frac{2}{3} \Rightarrow AD = 9$ cm

$\frac{AG}{AD} = \frac{2}{1} \Rightarrow \frac{6}{GD} = \frac{2}{1} \Rightarrow GD = 3$ cm

(b) $\frac{AG}{GD} = \frac{2}{1} \Rightarrow \frac{12}{GD} = \frac{2}{1} \Rightarrow GD = 6$ cm

$AD = AG + GD = 12 + 6 = 18$ cm

(c) $\frac{AG}{GD} = \frac{2}{1} \Rightarrow \frac{8}{GD} = \frac{2}{1} \Rightarrow GD = 4$ cm

$AD = AG + GD = 8 + 4 = 12$ cm

(d) $AG = 14$ cm

$\frac{AG}{GD} = \frac{2}{1}$

$\frac{14}{GD} = \frac{2}{1}$

$GD = 7$ cm

$AD = AG + GD = 14 + 7 = 21$ cm

WORKSHEET – 9

Level – I

1) A	2) A	3) B	4) B	5) B	6) C	7) A
8) C	9) B	10) A	11) B	12) D	13) A	14) A

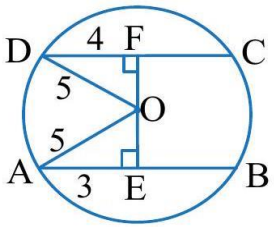
Level – III

1) A	2) B	3) B	4) D	5) A	6) B	7) A
8) D	9) D	10) D	11) A	12) C	13) C	14) A
15) A	16) A	17) B	18) B	19) A	20) B	21) B
22) A	23) B	24) A,C	25) A,C	26) D	27) A	28) B
29) 2	30) a-s;,b-r;c-q;d-p	31) A,B,D		32) D	33) B	34) A
35) a-s;b-p;c-q;d-r						

SOLUTIONS

1. $d = 2r = 2 \times 3 = 6$
2. $d = 2r = 2 \times 5 = 10$
3. $r = \frac{d}{2} = \frac{20}{2} = 10$
4. $e = \frac{d}{2} = \frac{15}{2} = 75$
5. Conceptual
6. Conceptual
7. Conceptual
8. Conceptual
9. Circumference of circle $= 2\pi r = \frac{2\pi d}{2} = \pi d \left[\because r = \frac{d}{2}\right]$ 10. $C = 2\pi r$
10. $A = \pi r^2$
12. $AC = \sqrt{OA^2 - OC^2} = \sqrt{25 - 9} = 4$
    $AB = 2(4) = 8$
13. $OA = \sqrt{OC^2 + AC^2} = \sqrt{16 + 9} = 5$
14. $OC = \sqrt{OA^2 - AC^2} = \sqrt{169 - 144} = 5$
15. $C = 2\pi r 44 = 2 \times \frac{22}{7} \times r$
16. $A = \pi r^2 \Rightarrow 154 = \frac{22}{7} \times r^2$
17. $AC = \sqrt{OA^2 - AC^2} = \sqrt{100 - 36} = 8$
18. $AC = \sqrt{OA^2 - OC^2} = \sqrt{225 - 144} = 9$
19. $A = \pi r^2 = \frac{22}{7} \times 7 \times 7 = 154$ cm^2
20. $C = \pi d = \frac{22}{7} \times 14 = 44$ cm
21. $OC = \sqrt{OA^2 - AC^2} = \sqrt{100 - 36} = 8$

22. $A = \pi r^2 \Rightarrow 154 = \frac{22}{7} \times r^2 \Rightarrow r = 7$



OF = 3 cm
OE = 4 cm
∴ EF = 7 cm

24. $C = 2\pi r = \pi d$
25. $A = 2\pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
26. Conceptual
27. Conceptual
28. Conceptual
29. $C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm
30. Conceptual.
31. $A = \pi r^2 = \pi \times 49 = \frac{22}{7} \times 49 = 154$ cm^2

$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44$ cm

$d = \frac{r}{2}$ cm

32. AC = 12 cm
33. $OC = \sqrt{OA^2 - AC^2} = \sqrt{37^2 - 12^2} = 35$
34. $d = 2r = 74$; r=37
35. A) $OA = \sqrt{OC^2 + AC^2} = \sqrt{9 + 16} = 5$

B) $OA = \sqrt{OC^2 + AC^2} = \sqrt{25 + 144} = 13$

C) $OA = \sqrt{OC^2 + AC^2} = \sqrt{49 + 576} = 25$

D) $OA = \sqrt{OC^2 + AC^2} = \sqrt{441 + 400} = 29$

9. TRIGONOMETRY

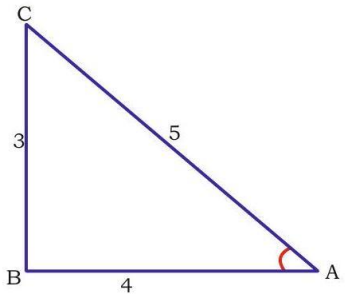
WORKSHEET – 1

Level – I

1.C	2.B	3.A	4.C	5.C	6.D	7.A
8.C	9.C	10.D	11.C			

SOLUTONS

1. Conceptual
2. Conceptual
3. Conceptual
4. Conceptual
5. Sol: $\frac{D}{90}=\frac{2C}{\pi}\Rightarrow\frac{D}{90}=\frac{2\cdot\frac{\pi}{6}}{\pi}\Rightarrow\frac{D}{90}=\frac{1}{3}\Rightarrow D=30°$
6. Sol: $\frac{D}{90}=\frac{G}{100}\Rightarrow\frac{330^0}{90}=\frac{G}{100}\Rightarrow\frac{11}{3}=\frac{G}{100}$
7. Conceptual
8. Sol: In $\triangle ABC, \underline{B}=90$



$\Rightarrow \text{Sin}\, A=\frac{3}{5}$

9. Conceptual
10. Conceptual
11. SOL: $\tan\theta\cdot\cot\theta=1\Rightarrow\frac{1}{\cot\theta}=\tan\theta\Rightarrow\frac{1}{\cot\theta}=\frac{\sin\theta}{\cos\theta}$

Level – II

1. B	2. D	3. A	4. C	5. C	6. D	7. A
8. C	9. B	10. A	11. B	12. C	13. C	14. B
15. B	16. A	17. B	18. A			

1. $\frac{D}{90}=\frac{G}{100}\Rightarrow\frac{330^0}{90}=\frac{G}{100}\Rightarrow\frac{11}{3}=\frac{G}{100}\Rightarrow G=\frac{1100^g}{3}$

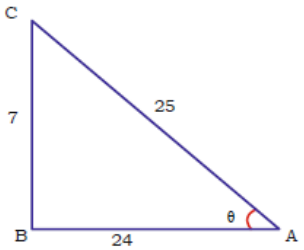
2. $\frac{D}{90}=\frac{2C}{\pi}\Rightarrow\frac{D}{90}=\frac{2(3\pi)}{\pi}\Rightarrow\frac{D}{90}=6\Rightarrow D=540°$

3. Sol: $\frac{D}{90}=\frac{G}{100}=\frac{2C}{\pi}\ \frac{D}{90}=\frac{200}{100}\Rightarrow\frac{D}{90}=\frac{2}{1}\Rightarrow D=180°$

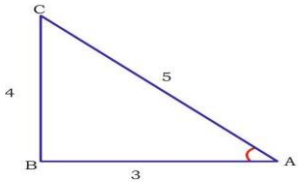
$\frac{200}{100}=\frac{2C}{\pi}\Rightarrow\frac{2}{1}=\frac{2C}{\pi}\Rightarrow C=\frac{2\pi}{2}\ \Rightarrow 200^{g}=\pi^{c}$

4. Sol:

From the triangle $\operatorname{cosec}A=\frac{hyp}{opp}=\frac{25}{7}$



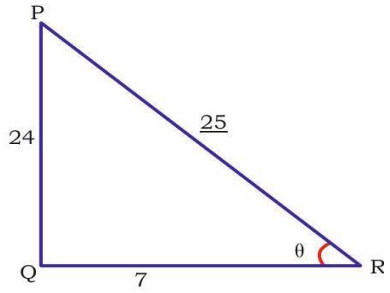
5.



$\cos A=\frac{adj}{hyp}=\frac{3}{5},\sec A=\frac{hyp}{adj}=\frac{5}{3}$

$\cos A\cdot\operatorname{cosec}A=\frac{3}{5}\cdot\frac{5}{3}=1$

6.



$3\sin rR+4\cos R=3\left(\frac{24}{25}\right)+4\left(\frac{7}{25}\right)=\frac{72}{25}+\frac{28}{25}=\frac{100}{25}=4$

7. $\sin^{2}\theta\cdot\operatorname{cosec}e^{2}\theta=(\sin\theta\cdot\operatorname{cosec}\theta)^{2}=1^{2}=1$

8) $\sin\theta+\cos\theta=\frac{4}{5}+\frac{3}{5}=\frac{7}{5}$ 9) $1^{g}=100^{1}$ 10) $\frac{D}{90}=\frac{G}{100}\Rightarrow\frac{D}{100}=\frac{800}{100}\Rightarrow D=720$

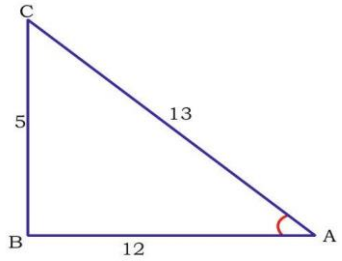
11. Sol: $\frac{D}{90}=\frac{2C}{\pi}\Rightarrow\frac{D}{90}=\frac{2\cdot\left(\frac{5\pi}{3}\right)}{\pi}\Rightarrow D=90\times\frac{10}{3}\Rightarrow D=300$

12. $\frac{D}{90}=\frac{2C}{\pi}\Rightarrow\frac{40}{90}=\frac{2c}{\pi}\Rightarrow\frac{2\pi}{9}=c$

13. Sol: $\sin A + \cos A = \frac{12}{13} + \frac{15}{13} = \frac{17}{13}$

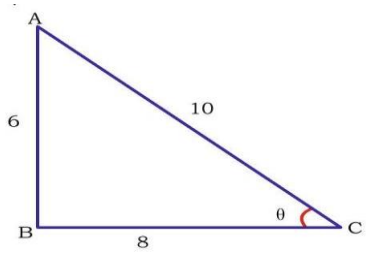
14. Sol: $\sin^4\theta - \sin^2\theta = \left(\frac{4}{5}\right)^4 - \left(\frac{4}{5}\right)^2 = \frac{256}{625} - \frac{16}{25} = \frac{256-400}{625} = \frac{-144}{625}$

15. Sol:



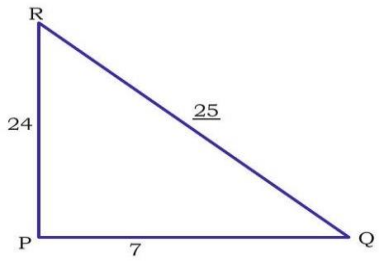
$\sin A\cos A + \sin C \cdot \cos C = \left(\frac{5}{13}\right)\left(\frac{12}{13}\right) + \left(\frac{12}{13}\right)\left(\frac{5}{13}\right) = \frac{120}{169}$

16. Sol: $\cos^2\theta - \sin^2\theta = \left(\frac{21}{29}\right)^2 - \left(\frac{20}{29}\right)^2 = \frac{441}{841} - \frac{400}{841} = \frac{441-400}{841} = \frac{41}{841}$



$\text{Tan}\,\theta = \frac{3}{8}k \Rightarrow \frac{6}{8} = \frac{3}{8}k \Rightarrow 6 = 3k \Rightarrow k = 2 \;\therefore \tan\theta = \frac{6}{8}$

8. Sol:



$\tan Q = \tan R = \frac{24}{7} - \frac{7}{24} = \frac{576-49}{168} = \frac{527}{168}$

Level – III

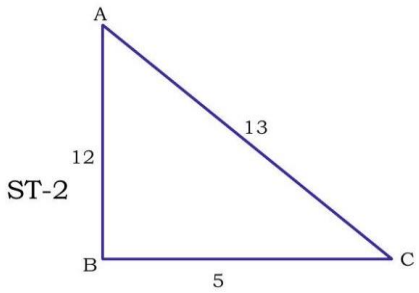
19. A,B,C,D 20. A 21. C 22. A 23. B

24. a-q; b-p,r; c-p,r; d-q 25.a-q; b-r; c-s; d-q 26.B,C,D

27. A,D 28. A 29. A 30. A 31. B 32. B

19. Sol: $\cos\theta \cdot \sec\theta = 1$

20. ST- 1: $\sin A + \cos A = \frac{12}{13} + \frac{5}{13} = \frac{17}{13}$



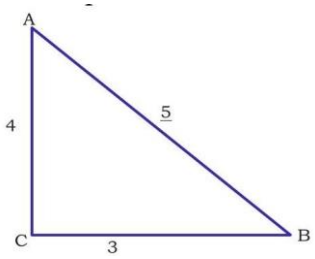
21. Sol: $\cos\theta + \sin\theta = \frac{5}{13} + \frac{12}{13} = \frac{17}{13}$

22. Sol: $\cos\theta - \sin\theta = \frac{5}{13} - \frac{12}{13} = \frac{7}{13}$

23. Sol: $\sec\theta \cdot \operatorname{cosec}\theta = \frac{13}{5} \cdot \frac{13}{12} = \frac{169}{60}$

24. Conceptual

25.



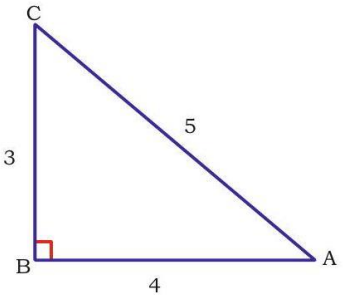
26. $\frac{\text{Tan}\,\theta}{1/\text{Tan}\,\theta} = \text{Tan}^2\,\theta = \frac{\text{Sin}^2\,\theta}{\text{Cos}^2\,\theta} = \text{Sin}^2\,\theta \cdot \text{Sec}^2\,\theta$

27. Sol: $\sin B + \cos B = \frac{5}{13} + \frac{12}{13} = \frac{17}{13}$

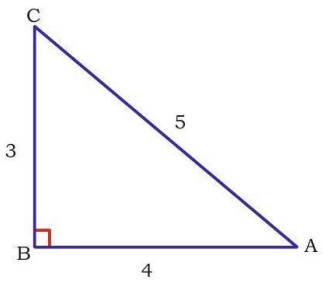
28. Sol: $\tan B = \frac{opp}{adj} = \frac{5}{12}$

29. Sol: $\sin^2 A + \cos^2 A = \left(\frac{5}{13}\right)^2 + \left(\frac{12}{13}\right)^2 = \frac{25}{169} + \frac{144}{169} = \frac{169}{169} = 1$

Sol:



31.

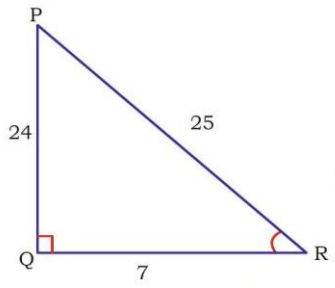


$$\sec^2\theta - \tan^2\theta = \left(\frac{5}{4}\right)^2 - \left(\frac{3}{4}\right)^2 = \frac{25}{16} - \frac{9}{16} = 1$$

$$\frac{1+\cot A}{1-\cot A} = \frac{1+\frac{4}{3}}{1-\frac{4}{3}} = \frac{\frac{7}{3}}{\frac{-1}{3}} = -7$$

32.

Sol:



$$1-\text{Cos}^2\theta = 1-\left(\frac{7}{25}\right)^2 = 1-\frac{49}{625} = \frac{576}{625}$$

WORKSHEET – 2

Level – I

1.B	2.B	3.A	4.C	5.D	6.C

Level – II

1.A	2.B	3.B	4.B	5.B	6.B	7.A
8.A	9.C	10.D	11.A	12.A	13.B	14.D
15.A	16.A	17.C	18.B	19.A	20.A	21.A

1. Ans: 1

   by Pythogoras theorm, (Hypotense)^2 = (side)^2 + (side)^2 $\Rightarrow (AC)^2 = (AB)^2 + (BC)^2$

2. Ans: $2\ \sin\theta = \frac{\text{opp side}}{\text{hyp}} = \frac{AB}{AC} = \frac{4}{5}$

   $(AC)^2 = (AB)^2 + (BC)^2$

   $\Rightarrow \sqrt{(AC)^2 - (AB)^2} = BC$

   $\Rightarrow \sqrt{(5)^2 - (4)^2} = BC \Rightarrow \sqrt{25-16} = BC \Rightarrow BC = 3 \Rightarrow \cos\theta = \frac{4}{5}$

3. Ans: 2

   $(AC)^2 = (AB)^2 + (BC)^2 \Rightarrow (13)^2 = (AB)^2 + (12)^2 \Rightarrow 169 = AB^2 + 144 \Rightarrow 25 = AB^2$

   $\Rightarrow AB = 5$

4. Ans: 2

   $\text{Sin}\,A = \frac{\text{opp side}}{\text{hyp}} = \frac{3}{5}$

5. Ans: $2\ \cos C = \frac{\text{Adj side}}{\text{hyp}} = \frac{3}{5}$

6. Ans: 2

$\sin\theta = \frac{\text{opp side}}{\text{hyp}} = \frac{4}{5}$, then $\cos\theta = \frac{\text{adj side}}{\text{hyp}} = \frac{3}{5}$

$\sec\theta = \frac{1}{\cos\theta} = \frac{5}{3}$

7. Ans: 1

$AC = 29, AB = 20$

$(AC)^2 = (AB)^2 + (BC)^2 \Rightarrow (29)^2 = (20)^2 + (BC)^2 \Rightarrow 841 = 400 + (BC)^2$

$\Rightarrow \sqrt{841 - 400} = BC \Rightarrow \sqrt{441} = BC \Rightarrow BC = 21$

8. $\text{Tan}\, R = \frac{\sin R}{\cos R} = \frac{\frac{PQ}{PR}}{\frac{QR}{PR}} = \frac{PQ}{QR} = \frac{8}{15}$

$(\text{ hyp })^2 = (s)^2 + (s)^2 \Rightarrow (\text{ hyp })^2 = (8)^2 + (15)^2 \Rightarrow (\text{ hyp })^2 = 64 + 225$

$\Rightarrow (\text{ hyp })^2 = 289 \Rightarrow \text{hyp} = 17 \sin R = \frac{PQ}{PR} = \frac{8}{17}$

9. $\Rightarrow (PR)^2 = (PQ)^2 + (QR)^2 \Rightarrow (PR)^2 = (15)^2 + (8)^2 \Rightarrow (PR)^2 = 289 \Rightarrow PR = 17$

$\sin\theta = \frac{\text{opp side}}{\text{hyp}} = \frac{15}{17}$

10. $3\sec\theta = 5 \Rightarrow \sec\theta = \frac{5}{3} \Rightarrow \cos\theta = \frac{1}{\sec\theta} \Rightarrow \cos\theta = \frac{3}{5}$

11. $3\sin\theta - 4\cos\theta = 0 \Rightarrow 3\sin\theta = 4\cos\theta$

$\Rightarrow \tan\theta = \frac{4}{3}$ then $\sin\theta = \frac{4}{5}, \cos\theta = \frac{3}{5}$

$\Rightarrow 3\sin\theta + 4\cos\theta = 3\left(\frac{4}{5}\right) + 4\left(\frac{3}{5}\right) = \frac{12}{5} + \frac{12}{5} = \frac{24}{5} = 4.8$

12. $(AC)^2 = (AB)^2 + (BC)^2 \Rightarrow (AC)^2 = (20)^2 + (15)^2 \Rightarrow (AC)^2 = 400 + 225$

$\Rightarrow AC = \sqrt{625} \Rightarrow AC = 25$

$\sin\theta = \frac{\text{opp side}}{\text{hyp}} = \frac{20}{25}, \cos\theta = \frac{\text{Adj side}}{\text{hyp}} = \frac{15}{25}$ $\sin\theta + \cos\theta = \frac{20}{25} + \frac{15}{25} \Rightarrow \frac{35}{25}$

13. Ans: 2 Sol: $\tan\theta = -\frac{3}{4}$ then $\sin\theta = ?$

$\frac{\sin\theta}{\cos\theta} = \tan\theta$ so $\sin\theta = \pm\frac{4}{5}$

$\tan\theta$ becomes positive in 3rd quadrant,

$\sin\theta$ is negative in 3rd &4th quadrant

then $\tan\theta$ is negative in 2nd &4th quadrant so $\sin\theta$ may be $+\frac{4}{5}$(or) $-\frac{4}{5}$

14. Ans: 4 $B = 90°$ $\tan A = 1 \Rightarrow A = 45°$ $\sin A = \sin 45 = \frac{1}{\sqrt{2}}$

15. $\tan\theta = \frac{12}{5} \Rightarrow \sin\theta = \frac{12}{13}; \cos\theta = \frac{5}{13}; \text{cosec}\,\theta = \frac{13}{12}$

$$\frac{\sin\theta+\cos\theta}{\cos\theta+\operatorname{cosec}\theta}=\frac{\frac{12}{13}+\frac{5}{13}}{\frac{5}{13}+\frac{13}{12}}=\frac{\frac{17}{13}}{\frac{60+169}{156}}=\frac{17}{13}\times\frac{156}{229}=\frac{204}{229}$$

16. $2\tan\theta=\sec\theta\Rightarrow\frac{\tan\theta}{\sec\theta}=\frac{1}{2}\Rightarrow\frac{\frac{\sin\theta}{\cos\theta}}{\frac{1}{\cos\theta}}=\sin\theta=\frac{1}{2}\Rightarrow\cos\theta=\frac{\sqrt{3}}{2}$

$\Rightarrow 4\cot^2\theta-3\tan^2\theta=4(3)-3\left(\frac{1}{3}\right)=12-1=11\ \Rightarrow 10+k=11\Rightarrow k=1$

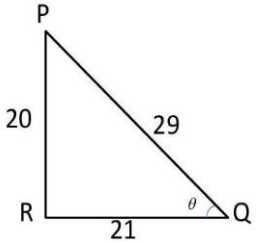
17. $\cot\theta=\frac{\cos\theta}{\sin\theta}=\frac{7}{8}$ By pythogoras theorm $\therefore\sin\theta=\frac{8}{\sqrt{113}}$

$(7)^2+(8)^2=(\text{ hyp })^2\ \cos\theta=\frac{7}{\sqrt{113}}\ 113=(\text{ hyp })^2$

$$\frac{(1+\sin\theta)(1-\sin\theta)}{(1+\cos\theta)(1-\cos\theta)}=\frac{\left(1+\frac{8}{\sqrt{113}}\right)\left(1-\frac{8}{\sqrt{113}}\right)}{\left(1+\frac{7}{\sqrt{113}}\right)\left(1-\frac{7}{\sqrt{113}}\right)}=\frac{1-\frac{8}{\sqrt{113}}+\frac{8}{\sqrt{113}}-\frac{64}{113}}{1-\frac{7}{\sqrt{113}}+\frac{7}{\sqrt{113}}-\frac{49}{113}}\Rightarrow\frac{1-\frac{64}{113}}{1-\frac{49}{113}}$$

$$=\frac{\frac{113-64}{113}}{\frac{113-49}{113}}=\frac{49}{64}$$

18. Ans: 2



by $usin\,g$ pythogoras theorm $PR=20\cos\theta=\frac{Adj}{Hyp}=\frac{21}{29},\sin\theta=\frac{Opp}{Hyp}=\frac{20}{29}\ \cos^2\theta-$

$\sin^2\theta=\left(\frac{21}{29}\right)^2-\left(\frac{20}{29}\right)^2=\frac{441}{841}-\frac{400}{841}=\frac{41}{841}$

19. $\tan\theta=\frac{\sin\theta}{\sin\theta}$ So by using pythogoras theorm hyp $=\sqrt{m^2+n^2}\ \sin\theta=\frac{m}{\sqrt{m^2+n^2}}$

20. $\sin\theta=\frac{7}{25}\Rightarrow\cos\theta=\frac{24}{25},\cot\theta=\frac{24}{7},\ \tan\theta=\frac{7}{24}$

$$\frac{7\cot\theta-24\tan\theta}{7\cot\theta+24\tan\theta}=\frac{7\times\frac{24}{7}-24\times\frac{7}{24}}{7\times\frac{24}{7}+24\times\frac{7}{24}}=\frac{24-7}{24+7}=\frac{17}{31}$$

21) $\tan\theta=\frac{7}{24}$

Level – III

22. A,B,C 23. A 24. D 25.A 26.C 27.49 28.C

29.A 30.C 31. A,B,C 32.A 33. a-s; b-p,r; c-q; d-t

22) Conceptual

23. $AB = 20$ cm, $BC = 15$ cm

hyp $= BC^2 + AB^2 \Rightarrow AC^2 = BC^2 + AB^2 \Rightarrow AC^2 = 225 + 400 \Rightarrow AC = 25$

$\text{Sin}\, A = \dfrac{Opp}{Hyp} = \dfrac{BC}{AC} = \dfrac{15}{25}$

24. Ans: 4 we know that always $\sin^2\theta + \cos^2\theta = 1$ so, $\sin^2\theta + \cos^2\theta = 1$

25. $\tan\theta = \frac{Opp}{Adj} = \frac{8}{15}$

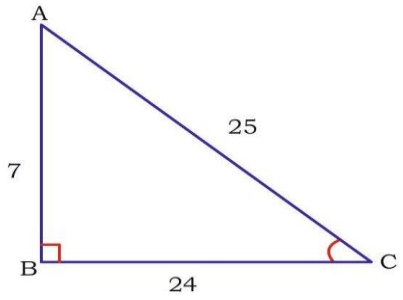
$Hyp^2 = (\text{ side })^2 + (\text{ side })^2 \Rightarrow Hyp^2 = (8)^2 + (15)^2 = 64 + 225 \Rightarrow 289 = Hyp^2$

$\Rightarrow Hyp = 17, Opp = 8, Adj = 15$

$\sec\theta = \dfrac{Hyp}{Adj} = \dfrac{17}{15}$

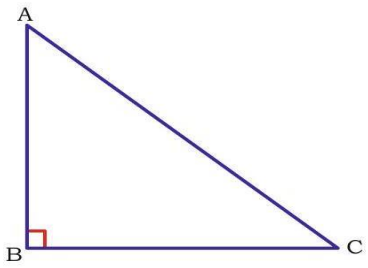
$15\sec\theta = 15 \times \dfrac{17}{15} \Rightarrow 15\sec\theta = 17^X$

26. $\sin\theta = \frac{Opp}{Hyp} = \frac{8}{17}$, $\cos\theta = \frac{Adj}{Hyp} = \frac{15}{17}$ $17\left(\frac{8}{17} + \frac{15}{17}\right) = 17\left(\frac{8+15}{17}\right) = 23$



$\text{cosec}\,\theta = \dfrac{25}{7}, \cos\theta = \dfrac{a}{b}$ $\cos\theta = \dfrac{Adj}{Hyp} = \dfrac{24}{25} = \dfrac{a}{b}$ a + b = 24 + 25 = 49

28. Ans: 3



$15\cos A - 8\sin A = 0$

Given, $15\cos A - 8\sin A = 0 \Rightarrow 15\cos A = 8\sin A$

$\Rightarrow \dfrac{\cos A}{\sin A} = \dfrac{8}{15} = \cot A$ then, $\sin A = \dfrac{15}{Hyp}; \cos A = \dfrac{8}{Hyp}$

$(Hyp)^2 = (\text{ side })^2 + (\text{ side })^2 \Rightarrow Hyp = \sqrt{(15)^2 + (8)^2} \Rightarrow Hyp = 17$

$\sin A = \dfrac{15}{17}, \cos A = \dfrac{8}{17}$ $\dfrac{\sin A + \cos A}{2\cos A - \sin A} = \dfrac{\frac{15}{17} + \frac{8}{17}}{2 \times \frac{8}{17} - \frac{15}{17}} \Rightarrow \dfrac{\frac{23}{17}}{\frac{16}{17} - \frac{15}{17}} \Rightarrow \dfrac{23}{17} \times \dfrac{17}{1} \Rightarrow 23$

29. we know that $\cot A = \frac{8}{15}$ $\sin A = \frac{15}{17} \Rightarrow \tan A = \frac{1}{\cot A} = \frac{1}{8/15} = \frac{15}{8}$, $\sec A = \frac{\sin A}{\tan A} = \frac{15/17}{15/8} =$

$\frac{15}{17} \times \frac{8}{15} = \frac{8}{17}$ $\frac{15\cot A + 17\sin A}{8\tan A + 16\sec A} = \frac{15\times\frac{8}{15} + 17\times\frac{15}{17}}{8\times\frac{15}{8} + 16\times\frac{17}{8}} = \frac{8+15}{15+34} = \frac{23}{49}$

30. $\operatorname{cosec} A = \frac{1}{\sin A} = \frac{1}{15/17} = \frac{17}{15}, \sec A = \frac{17}{8}$ $\frac{\operatorname{cosec} A - \sec A}{\operatorname{cosec} A + \sec A} = \frac{\frac{17}{15} - \frac{17}{8}}{\frac{17}{15} + \frac{17}{8}} = \frac{\frac{136-255}{120}}{\frac{136+255}{120}} = \frac{-119}{391}$

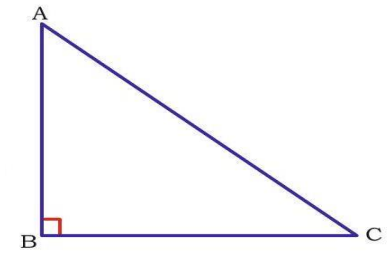
31. $\sin\theta = \frac{opp}{\text{hyp}} = \frac{p}{\sqrt{p^2+q^2}}$ then, $\cos\theta = \frac{q}{\sqrt{p^2+q^2}}$

$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{p}{\sqrt{p^2+q^2}} \times \frac{\sqrt{p^2+q^2}}{q} = \frac{p}{q}$ $\operatorname{cosec}\theta = \frac{1}{\sin\theta} = \frac{\frac{1}{p}}{\sqrt{p^2+q^2}} = \frac{\sqrt{p^2+q^2}}{p}$

32. Ans: 1

Sol:

: 1



$\tan\theta = \frac{1}{\sqrt{3}}$ then, $\cos A \cdot \cos C - \sin A \cdot \sin C = \cos(A + C)$

As, $\tan A = \frac{1}{\sqrt{3}} \Rightarrow A = 30^\circ \angle A + \angle B + \angle C = 180 \Rightarrow 30^\circ + 90^\circ + \angle C = 180^\circ \Rightarrow \angle C = 60^\circ$

So, $\cos A \cdot \cos C - \sin A \cdot \sin C = \cos(A + C)$

$\Rightarrow \cos A \cdot \cos C - \sin A \cdot \sin C = \cos(30^\circ + 60^\circ) = \cos(90^\circ)$

$\Rightarrow \cos A \cdot \cos C - \sin A \cdot \sin C = 0$

33. $\sec\theta = \frac{\text{hyp}}{\text{adj.side}}, \operatorname{cosec}\theta = \frac{\text{hyp}}{\text{opp.side}}, \sin\theta = \frac{\text{opp}}{\text{hyp}}, \cos\theta = \frac{\text{adj } j}{\text{hyp}}, \tan\theta = \frac{\text{opp.side}}{\text{adj.side}}$

WORKSHEET – 3

Level – I

1. B	2. D	3. D	4. A	5. B	6. A	7. B
8. B	9. D	10. B				

SOLUTIONS

1. 60°
2. $\left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}$
3. $1 + 1 + \sqrt{2}\left(\frac{1}{\sqrt{2}}\right) = 3$
4. $1 + 2(3) = 7$
5. $\cos 60^\circ + \tan 45^\circ = \frac{1}{2} + 1 = \frac{3}{2}$

6. $2(1) + 3 = 5$

7. $\frac{1}{2} - \frac{1}{2} = 0$

8. $4 + 4 = 8$

9. $\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{3}}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4} + \frac{1}{4} = 1$

10. $\frac{1}{2} \cdot \frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \cdot \frac{1}{2} = 0$

Level – II

1. D	2. C	3. A	4. D	5. A	6. B	7. A
8. A	9. D	10. B	11. A	12. A	13. C	14. B
15. B	16. A					

1. $\tan^2 45 + \cot^2 45 = 1^2 + 1^2 = 2$

2. $\sin 60° + \cos 30° = \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3}$

3. $\sin 0° + \sin 90° = 0 + 1 = 1$

4. $\sec^2 45 - \tan^2 45 = (\sqrt{2})^2 - 1^2 = 1$

5. $\frac{4}{3} + \frac{4}{3} + \frac{3}{2} = \frac{16+3}{6} = \frac{19}{6}$

6. $\frac{\sin^2 30 - \cos^2 30}{\sin^2 30 + \cos^2 30} = \frac{\frac{1}{4} - \frac{3}{4}}{\frac{1}{4} + \frac{3}{4}} = \frac{-2}{4} = \frac{-1}{2}$

7. $\frac{1 - \tan^2 60}{1 + \tan^2 60} = \frac{1-3}{1+3} = \frac{-2}{4} = \frac{-1}{2} = -\cos 60°$

8. $\cot^2 30 - 2\cos^2 30 - \frac{3}{4}\sec^2 45 + \frac{1}{4}\text{cosec}\, e^2 30 = 1 - 2 \cdot \frac{1}{4} - \frac{3}{4}(2) + \frac{1}{4}(4) = 1 - \frac{1}{2} - \frac{3}{2} + 1 = 0$

9. $\frac{\tan 60° - \tan 30°}{1 + \tan 60° \cdot \tan 30°} = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{1 + \sqrt{3} \cdot \frac{1}{\sqrt{3}}} = \frac{\frac{3-1}{\sqrt{3}}}{2} = \frac{1}{\sqrt{3}}$

10. $\frac{4}{\cot^2 30} + \frac{1}{\sin^2 30} - 2\cos^2 45 - \sin^2 0 = \frac{4}{3} + \frac{4}{1} - 2\left(\frac{1}{2}\right) - 0 = \frac{4}{3} + 3 = \frac{13}{3}$

11. $\frac{1}{\sqrt{2}} \cdot \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \cdot \frac{1}{2} = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$$\frac{\sin 30°}{1 - \cos 30°} - \frac{1 - \cos 30°}{\sin 30°} = \frac{\frac{1}{2}}{1 - \frac{\sqrt{3}}{2}} - \frac{1 - \frac{\sqrt{3}}{2}}{\frac{1}{2}} = \frac{\frac{1}{2}}{\frac{2-\sqrt{3}}{2}} - \frac{\frac{2-\sqrt{3}}{2}}{\frac{1}{2}} = \frac{1}{2-\sqrt{3}} - \frac{2-\sqrt{3}}{1}$$

$$= 2\sqrt{3}$$

13. $\frac{\cot 30°}{1 - \tan 30°} + \frac{\tan 30°}{1 - \cot 30°} = \frac{\sqrt{3}}{1 - \frac{1}{\sqrt{3}}} + \frac{\overline{\sqrt{3}}}{1 - \sqrt{3}} = \frac{3}{\sqrt{3}-1} + \frac{1}{\sqrt{3}-3}$

14. $\frac{4}{\cot^2 30°} + \frac{1}{\sin^2 30°} - 2\cos^2 45° - \sin^2 0° = \frac{4}{3} + 4 - 2\left(\frac{1}{2}\right) - 0 = \frac{13}{3}$

15. $\sqrt{3}\sin\theta - \cos\theta = 0 \Rightarrow \sqrt{3}\sin\theta = \cos\theta \Rightarrow \tan\theta = \frac{1}{\sqrt{3}}$

$$\frac{2\tan\theta}{1-\tan^2\theta}=\frac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\left(\frac{1}{\sqrt{3}}\right)^2}=\frac{\frac{2}{\sqrt{3}}}{\frac{3-1}{3}}=\frac{3}{\sqrt{3}}=\sqrt{3}$$

16.

$$\frac{\tan^2 60+4\cos^2 45+3\operatorname{cosec} e^2 60+2\cos^2 90}{2\operatorname{cosec} 30^\circ+3\sec 60^\circ-\frac{7}{3}\cot^2 30}=\frac{3+4\left(\frac{1}{2}\right)+3\left(\frac{4}{3}\right)+2(0)}{2(2)+3(2)-\frac{7}{3}\times 3}=\frac{9}{3}=3$$

Level – III

17. A,C,D	18. A	19. a-p,r,s;b-p,r,s;c-q;d-p,r,s		20. 1	21. C
22. A	23. B	24. C	25. a-q,r;b-s;c-p;d-s	26. 1	27. 3
28. B	29. C	30. B	31. a-s;b-p;c-t;d-r		

17. $\cos 90^\circ = 0 \therefore 0 \times 9 = 0$

18. ST-I $\frac{2\tan 45^\circ}{1+\tan^2 45^\circ}=\frac{2(1)}{1+1^2}=\frac{2}{2}=1$

ST-II $\frac{2\tan 30^\circ}{1+\tan^2 30^\circ}=\frac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\frac{1}{3}}=\frac{3}{\sqrt{3}}=\sqrt{3}$

19. Ans: a) $1, \tan 45^\circ, \cot 45^\circ \cdot \tan 45^\circ$ b) $1, \tan 45^\circ, \cot 45^\circ \cdot \tan 45^\circ$ c) $\operatorname{cosec} 0^\circ$ d) $1, \tan 45^\circ, \cot 45^\circ \cdot \tan 45^\circ$

20. $\sin 0^\circ + \cos 30^\circ - \tan 45^\circ + \sin 60^\circ = 0 + \frac{\sqrt{3}}{2} - 1 + \frac{\sqrt{3}}{2} = 1$

21. $\sin^2 60 + \cos^2 45 = \left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{\sqrt{2}}\right)^2 = \frac{3}{4}+\frac{1}{2}=\frac{3+2}{4}=\frac{5}{4}$

22. $\frac{\sin 30^\circ \cdot \cos 60^\circ}{\tan 45^\circ \cdot \cot 45^\circ}=\frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)}{1\cdot 1}=\frac{1}{4}$

23. $2\sin^2 30 - 3\cos^2 60 + \cot^2 30 = 2\left(\frac{1}{4}\right) - 3\left(\frac{1}{4}\right) + 3 = \frac{1}{2}-\frac{3}{4}+3=\frac{2-3+12}{4}=\frac{11}{4}$

24. $2\left(\frac{1}{2}\right)^2 - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$

$$1-2\left(\frac{\sqrt{3}}{2}\right)^2 = 1-\left(\frac{3}{4}\right) = -\frac{1}{2}$$

$$\left(\frac{1}{2}\right)^2 - \left(\frac{\sqrt{3}}{2}\right)^2 = \frac{1}{4}-\frac{3}{4}=-\frac{2}{4}=-\frac{1}{2}$$

$$2\left(\frac{\sqrt{3}}{2}\right)\left(\frac{1}{2}\right)=\frac{\sqrt{3}}{2}$$

25. a) $\operatorname{cosec}^2 30 + \cot^2 45 = 4 + 1 = 5$

$$\sin 30 + \cot 45 = \frac{1}{2}+1=\frac{3}{2}$$

b) $\frac{1-\sin 60^\circ}{\cos 60^\circ} = \frac{1-\frac{\sqrt{3}}{2}}{\frac{1}{2}} = 2-\sqrt{3}$

c) $\sin^2 60 - 3\cos^2 60 + 2\tan 45^\circ = \frac{3}{4} - 3\cdot\frac{1}{4} + 2 = 2$

d) $\frac{1-\tan 30^\circ}{1+\tan 30^\circ} = \frac{1-\frac{1}{\sqrt{3}}}{1+\frac{1}{\sqrt{3}}} = \frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(\sqrt{3}-1)^2}{2} = \frac{3+1-2\sqrt{3}}{2} = 2-\sqrt{3}$

26. $\cos 0^\circ + \sin 90^\circ - \sqrt{2}\cos 45^\circ = 1 + 1 - \sqrt{2}\cdot\frac{1}{\sqrt{2}} = 1+1-1 = 1$

27. $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ} = \frac{2\left(\frac{1}{\sqrt{3}}\right)}{1-\frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \sqrt{3}$

28. $\sin A + \cos B = \sqrt{3}$

29. $\frac{\sin 30}{1-\cos 30} + \frac{\sin 30}{1+\cos 30} = \frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}} + \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} = \frac{1}{2-\sqrt{3}} + \frac{1}{2+\sqrt{3}} = 4$

30. $\frac{\cos 60}{1-\sin 60} + \frac{\cos 60}{1+\sin 60} = \frac{\frac{1}{2}}{1-\frac{\sqrt{3}}{2}} + \frac{\frac{1}{2}}{1+\frac{\sqrt{3}}{2}} = \frac{1}{2-\sqrt{3}} + \frac{1}{2+\sqrt{3}} = 4$

31. a) $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ} = \frac{2\cdot\frac{1}{\sqrt{3}}}{1-\frac{1}{3}} = \frac{\frac{2}{\sqrt{3}}}{\frac{2}{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$

b) $\cos\frac{\pi}{3} + \tan\frac{\pi}{4} = \cos 60 + \tan 45 = \frac{1}{2} + 1 = \frac{3}{2}$

c) $\frac{\sqrt{3}}{2} + 1 - 1 + 1 - 1 = \frac{\sqrt{3}}{2}$

d) $\cos 0^\circ + \sin 90^\circ - \sqrt{2}\sin 45^\circ = 1 + 1 - 1 = 1$

WORKSHEET – 4

Level – I

1) A 2) C 3) C 4) A 5) B 6) C 7) A
8) B 9) C 10) C

SOLUTIONS

1. Ans: 1

   Sol: Trigonometric Identities

2. $\sin^2\theta + \cos^2\theta = 1 \Rightarrow \cos^2\theta = 1 - \sin^2\theta$

3. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec^2\theta - 1 = \tan^2\theta$

4. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow -\tan^2\theta - 1 = -\sec^2\theta \Rightarrow -(\tan^2\theta + 1) = -(\sec^2\theta)$

   $\Rightarrow \tan^2\theta + 1 = \sec^2\theta$

5. $\sin^2\theta + \cos^2\theta = 1 \Rightarrow \sin^2\theta = 1 - \cos^2\theta \Rightarrow \sin\theta = \pm\sqrt{1-\cos^2\theta}$

6. $\sec\theta + \tan\theta = p \rightarrow 1, \sec\theta + \tan\theta = x \rightarrow 2$

   Multiply 1&2

   $(\sec\theta + \tan\theta)(\sec\theta - \tan\theta) = p \times x \Rightarrow \sec^2\theta - \tan^2\theta = px$

$\Rightarrow 1 = px \Rightarrow x = \frac{1}{p}$

7. $\text{cosec}^{\,2}\theta - \cot^2\theta = 1 \Rightarrow \text{cosec}^2\,\theta = 1 + \cot^2\theta$

8. $\text{cosec}^{\,2}\theta - \cot^2\theta = 1 \Rightarrow -\cot^2\theta = 1 - \text{cosec}^2\,\theta$

$\Rightarrow \cot^2\theta = \text{cosec}\,e^2\theta - 1 \Rightarrow \cot\theta = \pm\sqrt{\text{cosec}^2\,\theta - 1}$

9. $\text{cosec}\,\theta - \cot\theta = k \to 1,\ \text{cosec}\,\theta + \cot\theta = x \to 2$

Multiply 1&2

$\Rightarrow (\text{cosec}\,\theta - \cot\theta)(\text{cosec}\,\theta + \cot\theta) = kx$

$\Rightarrow \text{cosec}\,e^2\theta - \cot^2\theta = kx \Rightarrow 1 - kx \Rightarrow x = \frac{1}{k}$

10. Identity $\sin^2\theta + \cos^2\theta = 1$

Level – II

1) B	2) C	3) C	4) C	5) A	6) A	7) C
8) D	9) C	10) B	11) D	12) C	13) D	14) B
15) A	16) A	17) B	18) D	19) A	20) C	

1. $\sin^2\theta + \cos^2\theta = 1 \Rightarrow a^2 + \cos^2\theta = 1 \Rightarrow \cos^2\theta = 1 - a^2 \Rightarrow \cos\theta = \pm\sqrt{1 - a^2}$

2. $(a+b)^2 = a^2 + b^2 + 2ab \Rightarrow \sin^2\theta + \cos^2\theta + 2\cos\theta\sin\theta.$

$(a-b)^2 = a^2 + b^2 - 2ab \Rightarrow \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta$

$\Rightarrow \sin^2\theta + \cos^2\theta + 2\cos\theta\sin\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 2$

$\Rightarrow 2 = (\sin\theta + \cos\theta)^2 + (\sin\theta - \cos\theta)^2$

3. $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \sin^2\theta + \cos^2\theta + 2\sin\theta\cos\theta + \sin^2\theta + \cos^2\theta - 2\sin\theta\cos\theta = 1 + 1 = 2$

4. $so, (\cos^2\theta)\sec^2\theta = \cos^2\theta \times \frac{1}{\cos^2\theta} = 1$

5. $(a+b)(a-b) = a^2 - b^2 \Rightarrow (1 - \sin\theta)(1 + \sin\theta) = (1 - \sin^2\theta)\ (1 + \tan^2\theta)(1 - \sin^2\theta) = \sec^2\theta \cdot \text{cosec}^2\,\theta = 1$

6. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec^2\theta - 1 = \tan^2\theta\ \Rightarrow (\tan^2\theta)(\cot^2\theta) = \tan^2\theta \times \frac{1}{\tan^2\theta} = 1$

7. $\sec^2\theta - \tan^2\theta = 1 \Rightarrow \sec^2\theta = 1 + \tan^2\theta \ldots\ldots$

$\text{cosec}^2\,\theta - \cot^2\theta = 1 \Rightarrow \text{cosec}^2\,\theta = 1 + \cot^2\theta$ ..... (2) $\frac{1}{\sec^2\theta} + \frac{1}{\text{cosec}^2\,\theta} \Rightarrow \cos^2\theta + \sin^2\theta = 1$

8. $\sec^2\theta = 1 + \tan^2\theta$ ... ... (1)

$\text{cosec}\,e^2\theta = 1 + \cot^2\theta.$

$\frac{\tan^2\theta}{\sec^2\theta} + \frac{\cot^2\theta}{\text{cosec}^2\,\theta} = \frac{\sin^2\theta}{\cos^2\theta} \times \cos^2\theta + \frac{\cos^2\theta}{\sin^2\theta} \times \sin^2\theta = \sin^2\theta + \cos^2\theta = 1$

9. $\frac{(1+\sin\theta)(1+\sin\theta)+(\cos\theta)(\cos\theta)}{(\cos\theta)(1+\sin\theta)} = \frac{(1+\sin^2\theta+2\sin\theta)+\cos^2\theta}{(\cos\theta)(1+\sin\theta)} = \frac{2+2\sin\theta}{(\cos\theta)(1+\sin\theta)}$

$\Rightarrow \frac{2(1+\sin\theta)}{(\cos\theta)(1+\sin\theta)} = \frac{2}{\cos\theta} = 2\sec\theta$

10.

$\sin\theta + \operatorname{cosec}\theta = 2$ S.O.B.S $\Rightarrow (\sin\theta + \operatorname{cosec}\theta)^2 = (2)^2$

$\Rightarrow \left(\sin^2\theta + \operatorname{cosec}^2\theta + 2 \times \sin\theta \times \frac{1}{\sin\theta}\right) = 4$

$\Rightarrow \sin^2\theta + \operatorname{cosec}^2\theta = 4 - 2 = \sin^2\theta + \operatorname{cosec}^2\theta = 2$

11. $\operatorname{cosec}\theta = \sqrt{n+1}$ S.O.B.S

$\Rightarrow \operatorname{cosec}\ ^2\theta = n+1 \Rightarrow \frac{1}{\sin^2\theta} = \frac{n+1}{1}\ \sin^2\theta = \frac{1}{n+1}.$

$\cos\theta = x$

S.O.B.S

$\Rightarrow \cos^2\theta = x^2$

$\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \left(\frac{1}{n+1}\right) + x^2 = 1 \Rightarrow x^2 = 1 - \left(\frac{1}{n+1}\right)$

$\Rightarrow \cos^2\theta = x^2 = \frac{(n+1)-1}{(n+1)} \Rightarrow \cos^2\theta = x^2 = \frac{n}{n+1} \Rightarrow \cos\theta = \sqrt{\frac{n}{n+1}}$

12. $1 + \tan\theta + \sec\theta + \cot\theta + \cot\theta\tan\theta + \cot\theta\sec\theta - \operatorname{cosec}\theta - \operatorname{cosec}\theta\tan\theta - \operatorname{cosec}\theta\sec\theta$

$= 2 + \tan\theta + \sec\theta + \cot\theta + \cot\theta\sec\theta - \operatorname{cosec}\theta\tan\theta - \operatorname{cosec}\theta\sec\theta - \operatorname{cosec}\theta$

$= 2 + \frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta} + \frac{\cos\theta}{\sin\theta} \times \frac{1}{\cos\theta} - \frac{1}{\sin\theta} \times \frac{\sin\theta}{\cos\theta} - \frac{1}{\sin\theta} \times \frac{1}{\cos\theta} + \frac{1}{\cos\theta} - \frac{1}{\sin\theta}$

$= 2 + \frac{\sin^2\theta + \cos^2\theta}{\sin\theta\cos\theta} - \frac{1}{\sin\theta\cos\theta} = 2 + 0 = 2$

13. $\cos\theta - 4\sin\theta = 1$

let $\sin\theta + 4\cos\theta = k.$

$(1) \times 4 \Rightarrow 4\cos\theta - 16\sin\theta = 4$

$(3) - (2) \Rightarrow -17\sin\theta = 4 - k \Rightarrow \sin\theta = \frac{k-4}{17}$

From (1), $\cos\theta - 4\sin\theta = 1$

$\Rightarrow \cos\theta - 4\left(\frac{k-4}{17}\right) = 1 \Rightarrow \cos\theta = \frac{1}{1} + 4\left(\frac{k-4}{17}\right) \Rightarrow \cos\theta = \frac{17+4k-16}{17} = \frac{4k+1}{17}$

$\therefore \sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \left(\frac{k-4}{17}\right)^2 + \left(\frac{4k+1}{17}\right)^2 = 1 \Rightarrow (k-4)^2 + (4k+1)^2 = 289\ k = \pm 4$

14. $\sec\theta = a + \frac{1}{4a}$

$\sec\theta + \tan\theta = a + \frac{1}{4a} + \sqrt{\left(a + \frac{1}{4a}\right)^2 - 1} = a + \frac{1}{4a} + \sqrt{\frac{(4a^2+1)^2 - (4a)^2}{(4a)^2}}$

$= a + \frac{1}{4a} + \sqrt{\frac{(4a^2+1)^2 - 4.4a^2}{4a}} = a + \frac{1}{4a} + \frac{\sqrt{(4a^2-1)^2}}{4a} = a + \frac{1}{4a} + \frac{4a^2-1}{4a}$

$= a + \frac{1}{4a} + a - \frac{1}{4a} = 2a$

15. $2\sec^2\theta - \sec^4\theta - 2\text{cosec}^2\theta + \text{cosec}^4\theta$

$= \sec^2\theta(2 - \sec^2\theta) - \text{cosec}\ ^2\theta(2 - \text{cosec}\, e^2\theta)$

$= (1 + \tan^2\theta)(2 - 1 - \tan^2\theta) - (1 + \cot^2\theta)(2 - 1 - \cot^2\theta)$

$= (1 + \tan^2\theta)(1 - \tan^2\theta) - (1 + \cot^2\theta)(1 - \cot^2\theta)$

$= 1 - \tan^4\theta - 1 + \cot^4\theta$

$= \cot^4\theta - \tan^4\theta$

16. $\sec\theta + \tan\theta = p$

$\sec\theta - \tan\theta = \frac{1}{p}$

$(1) + (2) \Rightarrow 2\sec\theta = \frac{p^2+1}{p}$

$(1) - (2) \Rightarrow 2\tan\theta = \frac{p^2-1}{p}$.....

$\frac{(4)}{(3)} \Rightarrow \frac{2\tan\theta}{2\sec\theta} = \frac{\frac{p^2-1}{p}}{\frac{p^2+1}{p}} \Rightarrow \sin\theta = \frac{p^2-1}{p^2+1}$

17. $\sqrt{\frac{1+\sin A}{1-\sin A}} = \sqrt{\frac{(1+\sin A)^2}{(1+\sin A)(1-\sin A)}} = \frac{\sqrt{(1+\sin A)^2}}{\sqrt{1-\sin^2 A}} = \frac{\sqrt{(1+\sin A)^2}}{\sqrt{\cos^2 A}} = \frac{1+\sin A}{\cos A}$

$= \sec A + \tan A$

18.

$x = \cot\theta + \cos\theta; y = \cot\theta - \cos\theta\ (x^2 - y^2)^2 = kxy$

$\Rightarrow [(\cot\theta + \cos\theta)^2 - (\cot\theta - \cos\theta)^2]^2 = kxy \Rightarrow (2\cot\theta \cdot \cos\theta)^2 = k(\cot^2\theta - \cos^2\theta)$

$\Rightarrow 4\cot^2\theta \cdot \cos^2\theta = k\cot^2\theta \cdot \cos^2\theta \Rightarrow k = 4$

19) $x = a\cos^3\theta, y = a\sin^3\theta$

$x^{2/3} + y^{2/3} = (a\cos^3\theta)^{2/3} + (a\sin^3\theta)^{2/3} = a^{2/3}(1) = a^{2/3}$

20.

$$\sqrt{\frac{\sec\theta-1}{\sec\theta+1}}+\sqrt{\frac{\sec\theta+1}{\sec\theta-1}}=\frac{\sqrt{(\sec\theta-1)^2}+\sqrt{(\sec\theta+1)^2}}{\sqrt{(\sec\theta+1)(\sec\theta-1)}}=\frac{\sec\theta-1+\sec\theta+1}{\sqrt{\sec^2\theta-1}}$$

$$=\frac{2\sec\theta}{\sqrt{\tan^2\theta}}$$

$$=2\operatorname{cosec}\theta$$

Level – III

21) A,B 22) a-t;b-q,s;c-p,r;d-q,s 23) D 24) C 25) D

26) A 27) 2 28) D 29) A 30) A 31) a-r,s;b-q;c-t;d-p

32) 3 33) A 34) C 35) D

21. Ans: 1,2

    Sol: Conceptual

22. Ans: a-t: b-q,s:c-p,r: d-q,s

23. $\frac{\sec\theta-\tan\theta}{a}=\frac{\frac{1}{a}}{a}=\frac{1}{a^2}$

24. $\tan\theta-\sec\theta=-\frac{1}{a}$

25. $(\sec\theta+\tan\theta)(\sec\theta-\tan\theta)=a\left(\frac{1}{a}\right)=1$

26. Conceptual

27. $(\sin\theta+\cos\theta)^2+(\sin\theta-\cos\theta)^2=2(\sin^2\theta+\cos^2\theta)=2$

28. $\sin A+\cos A=\sqrt{2},\tan A+\cot A=2$

    $\Rightarrow\frac{\sin A}{\cos A}+\frac{\cos A}{\sin A}=2\Rightarrow\frac{\sin^2 A+\cos^2 A}{\sin A\cdot\cos A}=2\Rightarrow\sin A\cdot\cos A=\frac{1}{2}$

29. $\sec A\cdot\operatorname{cosec}A=\frac{1}{\cos A}\cdot\frac{1}{\sin A}=\frac{1}{1/2}=2$

30. $\sin^2 A-\cos^2 A=(\sin A+\cos A)(\sin A-\cos A)$

    $\Rightarrow\sin A-\cos A=\sqrt{(\sin A+\cos A)^2-4\sin A\cos A}$

    $\Rightarrow\sin A-\cos A=\sqrt{(\sqrt{2})^2-4(1/2)}\Rightarrow\sin A-\cos A=\sqrt{2-2}\Rightarrow\sin A-\cos A=0$

31.

    a) $\sin^4 A-\cos^4 A=(\sin^2 A+\cos^2 A)(\sin^2 A-\cos^2 A)=-(\cos^2 A-\sin^2 A)=-\cos 2A$

    $=-\cos 2A$

    b) $1-\frac{\cos^2 A}{1+\sin A}=1-\frac{\cos^2 A(-\sin A+1)}{-\sin^2 A+1}$

    c) $1-\frac{\cos^2 A(1-\sin A)}{\cos^2 A}=\sin A$

    d) $\tan^2 A+\tan^4 A=\tan^2 A(1+\tan^2 A)=(\sec^2 A-1)\sec^2 A=\sec^4 A-\sec^2 A$

32. $3(\sin x+\cos x)^4+6(\sin x-\cos x)^2+4(\sin^6 x+\cos^6 x)$.

$(\sin x+\cos x)^4=(1+2\sin x\cos x)^2=1+4\sin^2 x\cos^2 x+4\sin x\cdot\cos x$

$\sin^6 x+\cos^6 x=(\sin^2 x+\cos^2 x)^3-3\sin^2 x\cos^2 x(\sin^2 x+\cos^2 x)=1-3\sin^2 x\cos^2 x$

$G.E=3(1+4\sin^2 x\cos^2 x)+4(1-3\sin^2 x\cos^2 x)=13$

33.

$$(1+\tan\theta+\sec\theta)(1+\cos^2\theta-\operatorname{cosec}\theta)=\left(1+\frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}\right)\left(1\times\frac{\cos\theta}{\sin\theta}-\frac{1}{\sin\theta}\right)$$

$$=\left(\frac{\cos\theta+\sin\theta+1}{\cos\theta}\right)\left(\frac{\sin\theta+\cos\theta-1}{\sin\theta}\right)=\frac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta}$$

$$=\frac{\sin^2\theta+\cos^2\theta+2\sin\theta\cos\theta-1}{\sin\theta\cos\theta}=\frac{2\sin\theta\cos\theta}{\sin\theta\cos\theta}=2$$

34. $\operatorname{cosec}A+\cot A=k\ldots\ldots(1),\operatorname{cosec}A-\cot A=\frac{1}{k}$

$$(1)+(2)\Rightarrow\operatorname{cosec}A=\frac{k^2+1}{2k}\Rightarrow 4x+\frac{1}{16x}=\frac{k^2+1}{2k}\Rightarrow 4x+\frac{1}{16x}=\frac{1}{2}\left(k+\frac{1}{k}\right)\Rightarrow k$$

$$=8x \text{ or } \frac{1}{8x}$$

35. $(\operatorname{cosec}\theta-\sin\theta)(\sec\theta-\cos\theta)(\tan\theta+\cot\theta)=\left[\frac{1}{\sin\theta}-\sin\theta\right]\left[\frac{1}{\cos\theta}-\cos\theta\right]\left[\frac{\sin\theta}{\cos\theta}+\frac{\cos\theta}{\sin\theta}\right]$

$=\left(\frac{1-\sin^2\theta}{\sin\theta}\right)\left(\frac{1-\cos^2\theta}{\cos\theta}\right)\left(\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cdot\cos\theta}\right)=\frac{\cos^2\theta}{\sin\theta}\cdot\frac{\sin^2\theta}{\cos\theta}\cdot\frac{1}{\sin\theta\cos\theta}=1$